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Padgett et al.

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(54) **POLYMER-BASED CARTRIDGE CASING FOR SUBSONIC AMMUNITION**

(71) Applicant: **PCP Tactical, LLC**, Sebastian, FL (US)

(72) Inventors: **Charles Padgett**, Sebastian, FL (US);
Robert Padgett, Sebastian, FL (US)

(73) Assignee: **PCP Tactical, LLC**, Sebastian, FL (US)

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(51) **Int. Cl.**
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(52) **U.S. Cl.**
CPC **F42B 5/307** (2013.01); **F42B 5/313** (2013.01); **F42B 8/04** (2013.01); **F42B 12/76** (2013.01); **F42B 33/02** (2013.01)

(58) **Field of Classification Search**
CPC F42B 5/30; F42B 5/307; F42B 5/32; F42B 5/34; F42B 5/28; F42B 5/313; F42B 8/02; F42B 8/04; F42B 8/10
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File history of U.S. Appl. No. 61/456,664, which corresponds to US 2012/0111219.

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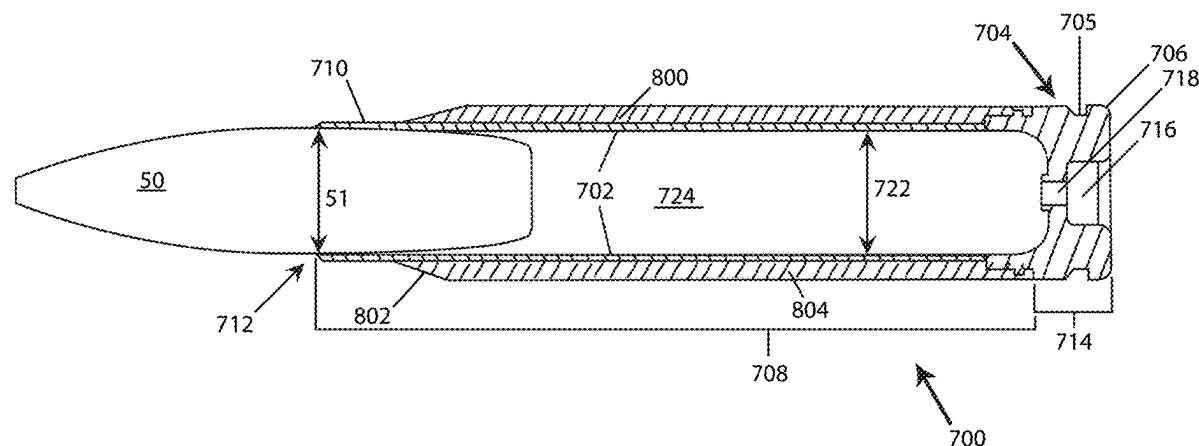
Primary Examiner — Joshua E Freeman

(74) *Attorney, Agent, or Firm* — Troutman Pepper Locke LLP

(57) **ABSTRACT**

A high strength polymer-based blank cartridge having a first end; a shoulder disposed approximate the first end; a second end opposite the first end; and an insert engaged to the second end. A metal sleeve extends between the shoulder and insert, and a polymer sheath covers at least the metal sleeve. A propellant chamber is formed between the shoulder and the insert by at least the metal sleeve, and the propellant chamber has an unobstructed and a reduced volume, wherein the reduced volume is reduced in comparison to a propellant chamber of a standard cartridge.

7 Claims, 19 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/063,051, filed on Oct. 5, 2020, now Pat. No. 11,353,299, which is a continuation of application No. 16/518,453, filed on Jul. 22, 2019, now Pat. No. 10,794,671, which is a continuation of application No. 16/257,262, filed on Jan. 25, 2019, now Pat. No. 10,359,263, which is a continuation of application No. 15/964,911, filed on Apr. 27, 2018, now Pat. No. 10,197,366, which is a continuation-in-part of application No. 15/187,421, filed on Jun. 20, 2016, now Pat. No. 9,995,561, which is a continuation of application No. 14/642,922, filed on Mar. 10, 2015, now Pat. No. 9,372,054, which is a continuation of application No. 14/315,564, filed on Jun. 26, 2014, now Pat. No. 9,003,973, said application No. 15/964,911 is a continuation-in-part of application No. 13/828,311, filed on Mar. 14, 2013, now abandoned, said application No. 14/315,564 is a division of application No. 13/549,351, filed on Jul. 13, 2012, now Pat. No. 8,763,535, which is a continuation-in-part of application No. 13/350,585, filed on Jan. 13, 2012, now abandoned, said application No. 13/828,311 is a continuation-in-part of application No. 13/350,585, filed on Jan. 13, 2012, now abandoned.

- (60) Provisional application No. 61/433,170, filed on Jan. 14, 2011.

(51) **Int. Cl.**

F42B 8/04 (2006.01)

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(58) **Field of Classification Search**

USPC 102/439

See application file for complete search history.

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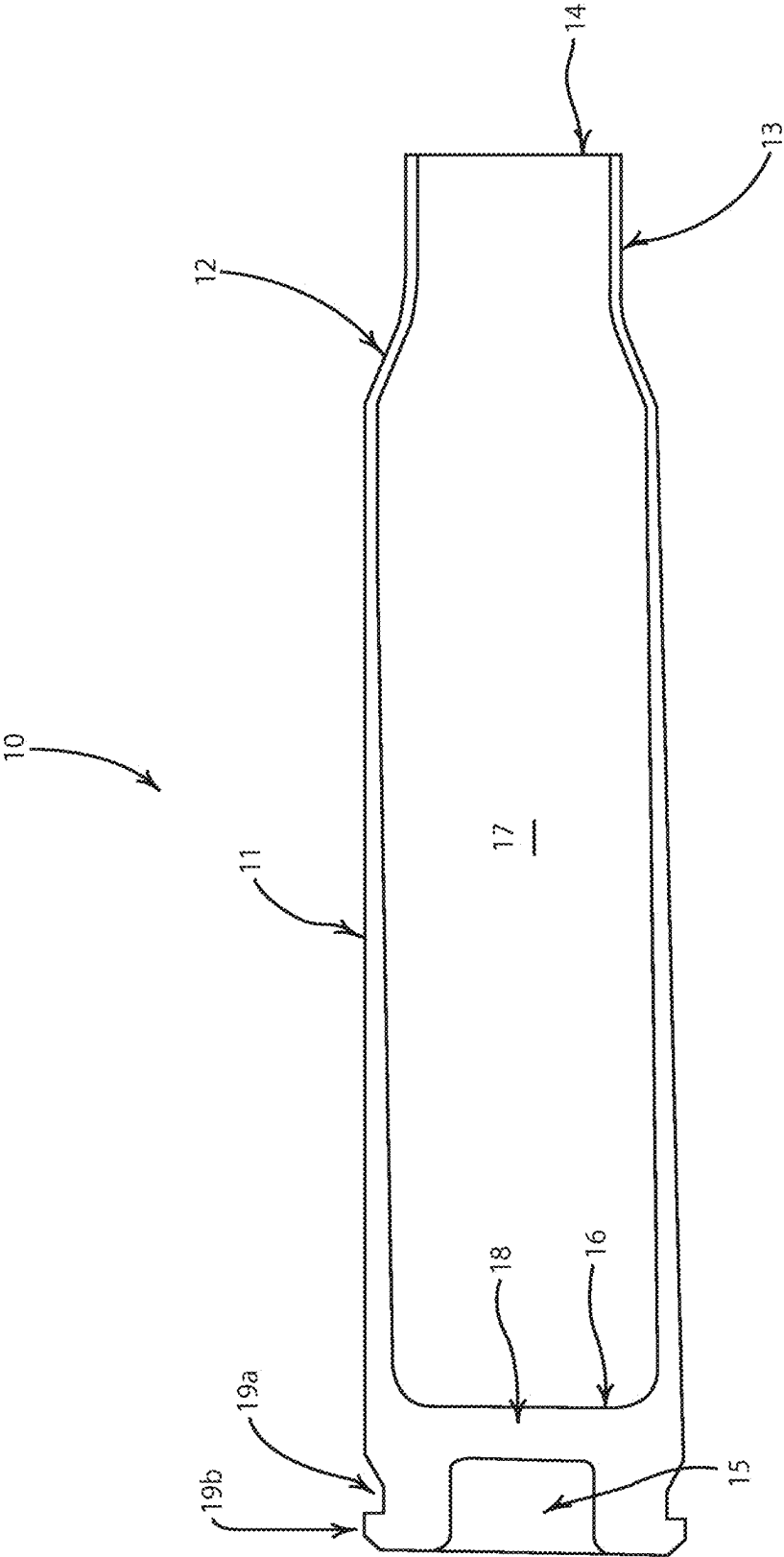


Fig. 1A
Prior Art

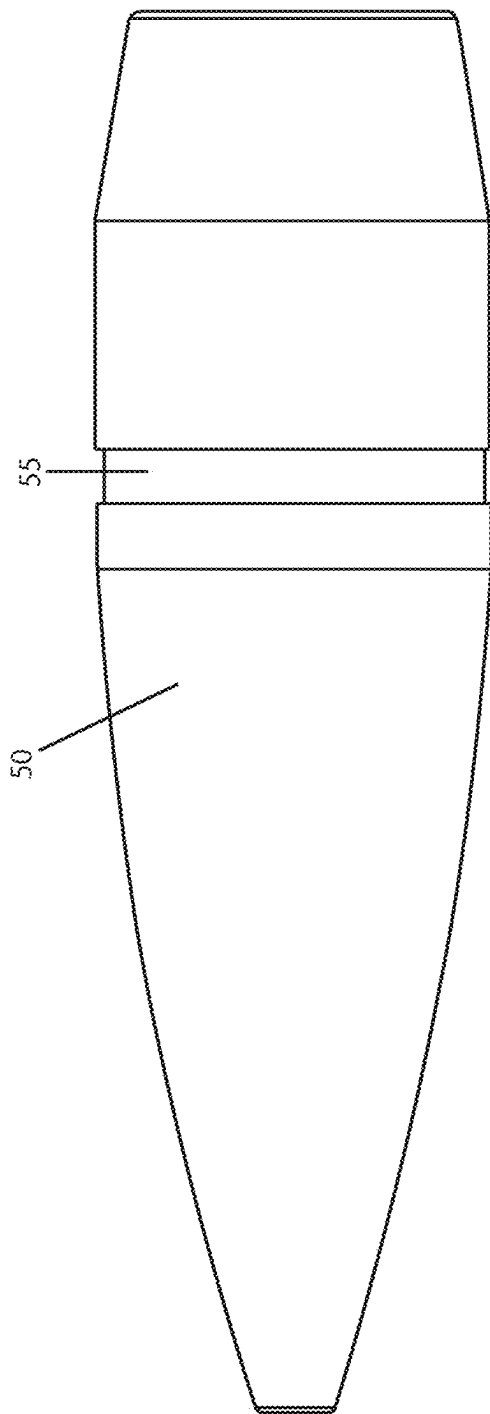


Fig. 1B
Prior Art

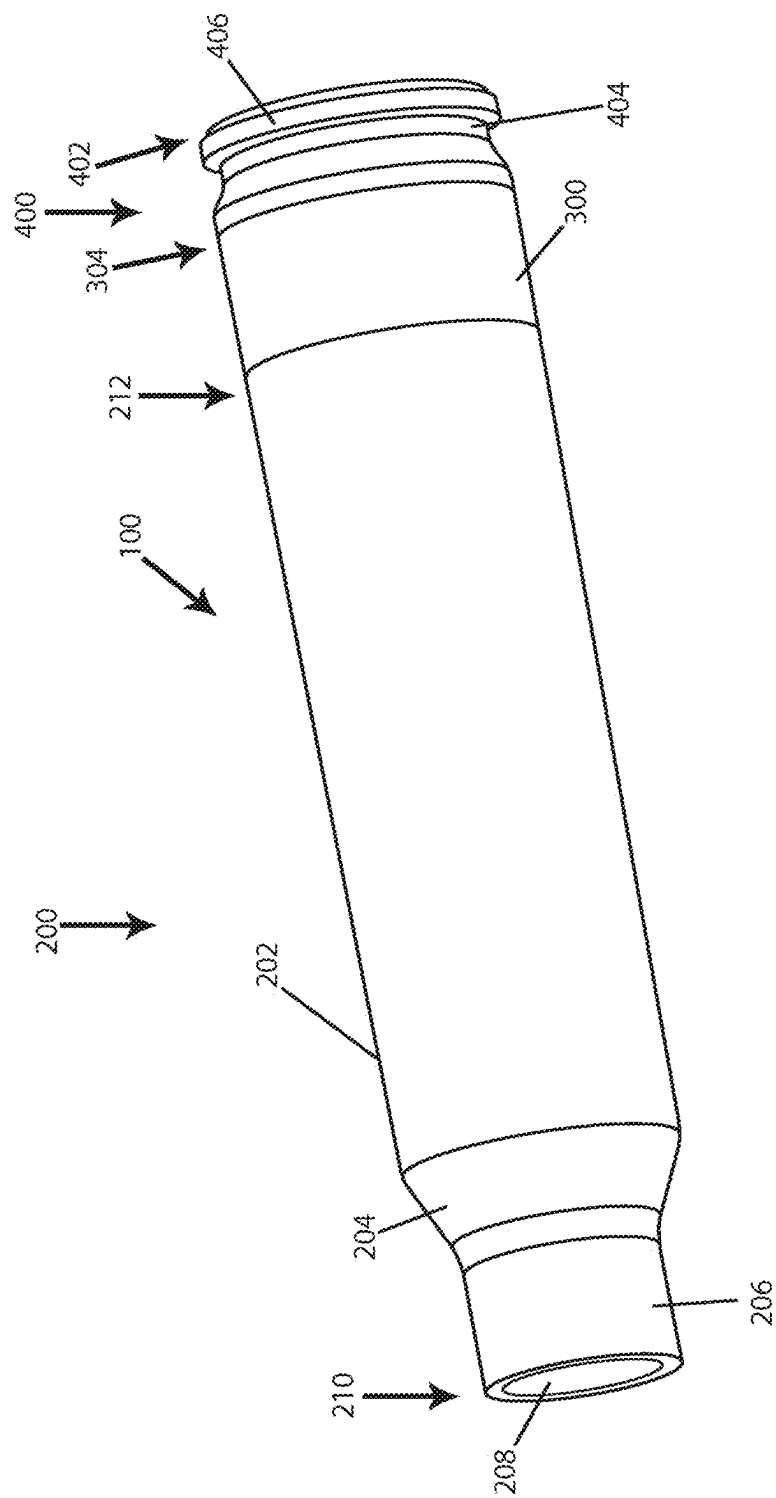


Fig.2

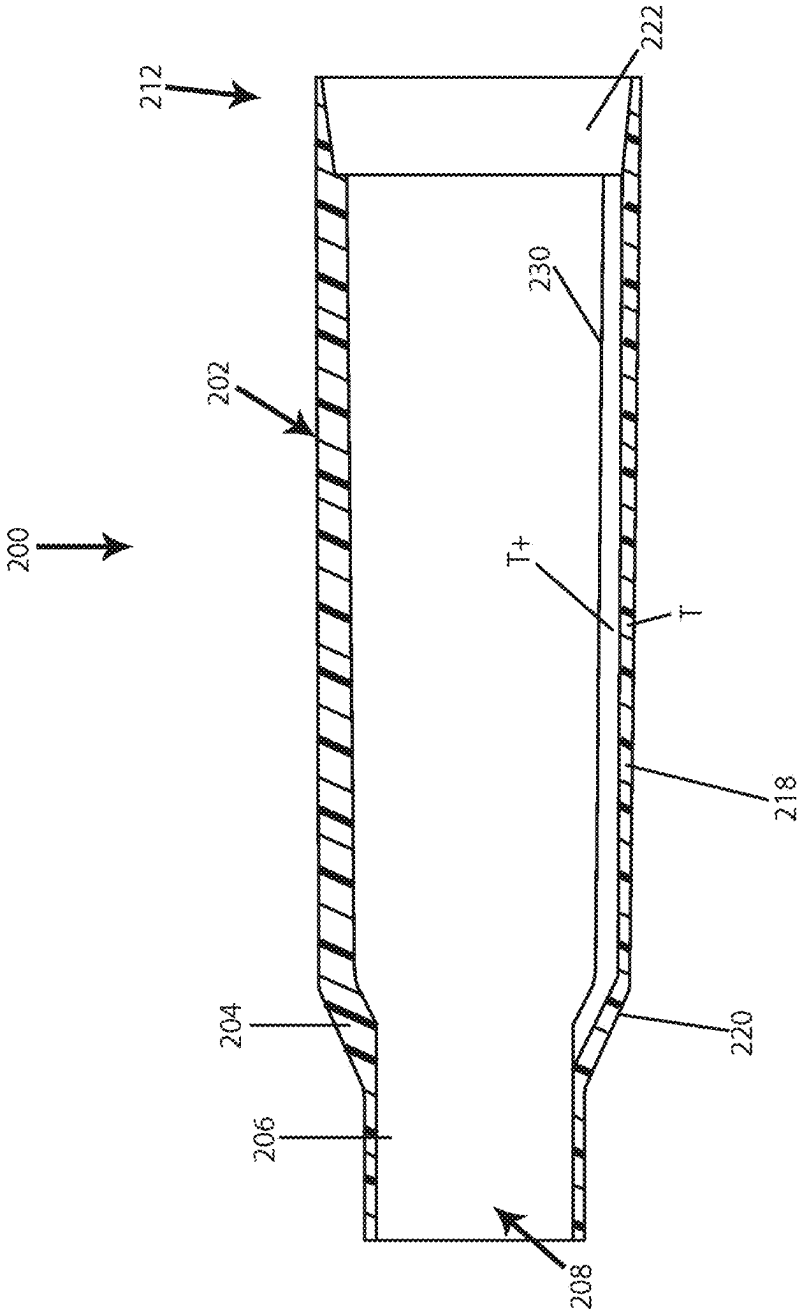


Fig. 3

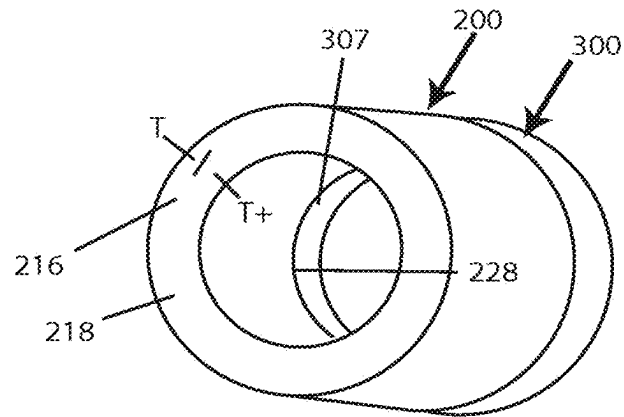


Fig. 4

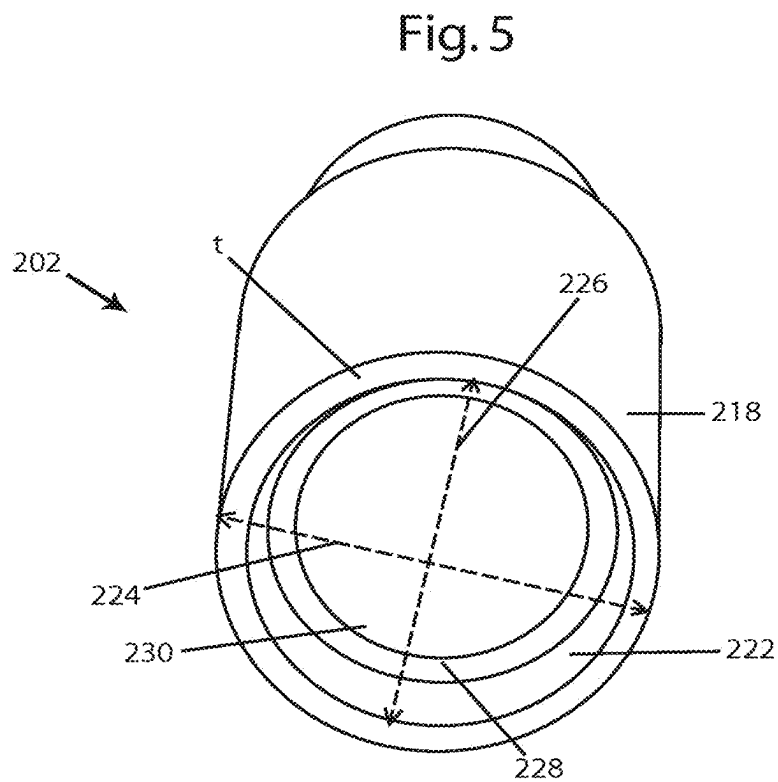


Fig. 5

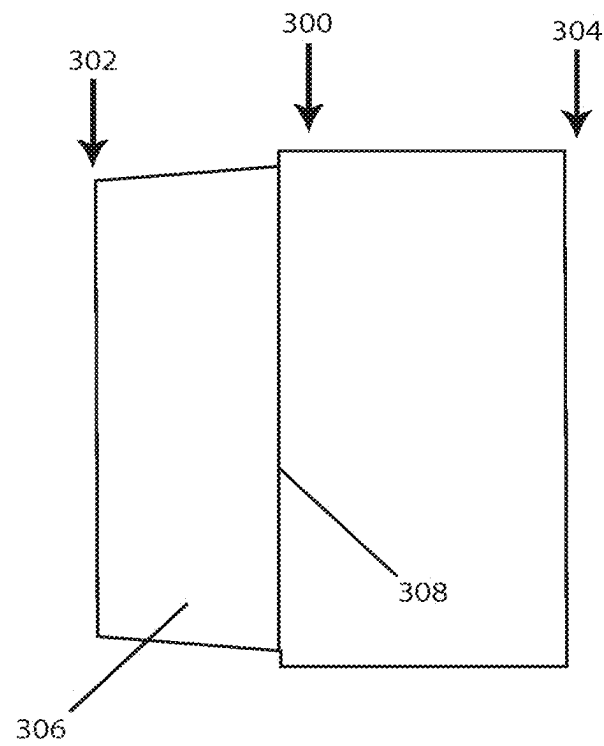


Fig. 6

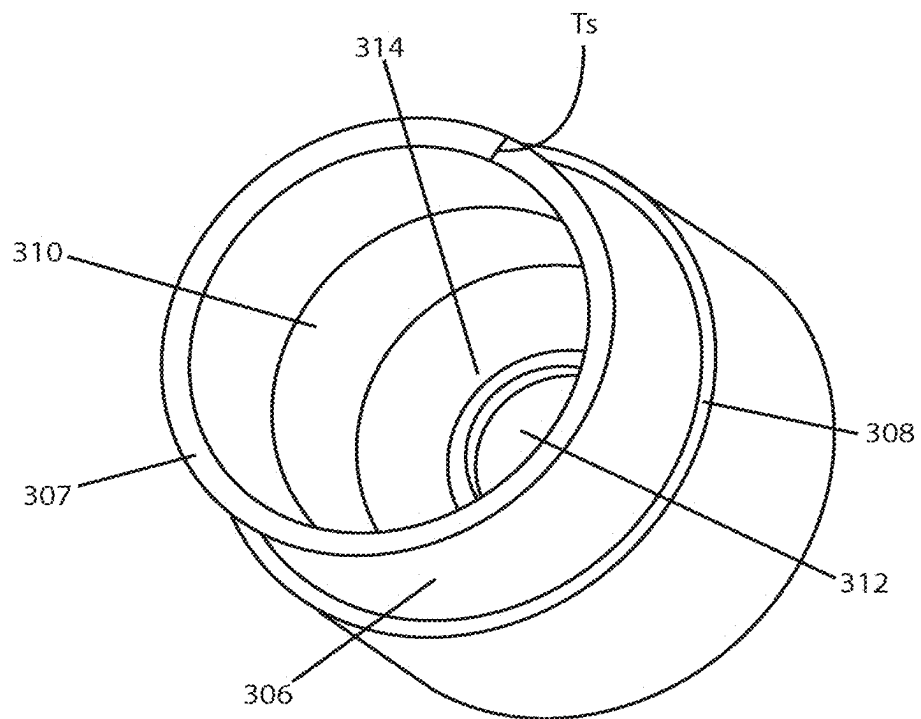


Fig. 7

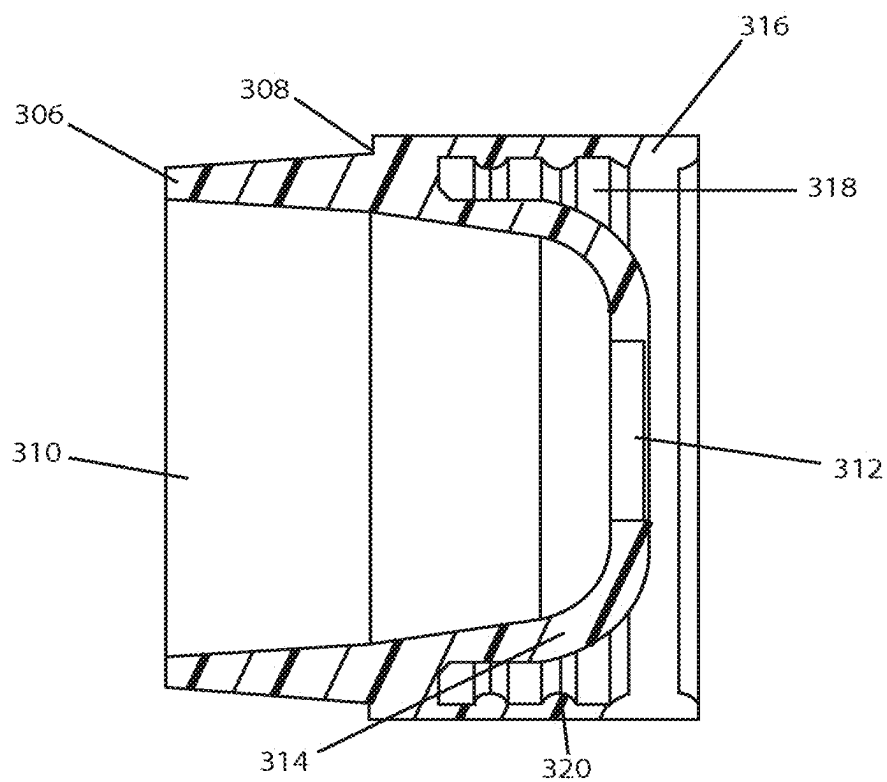


Fig. 8

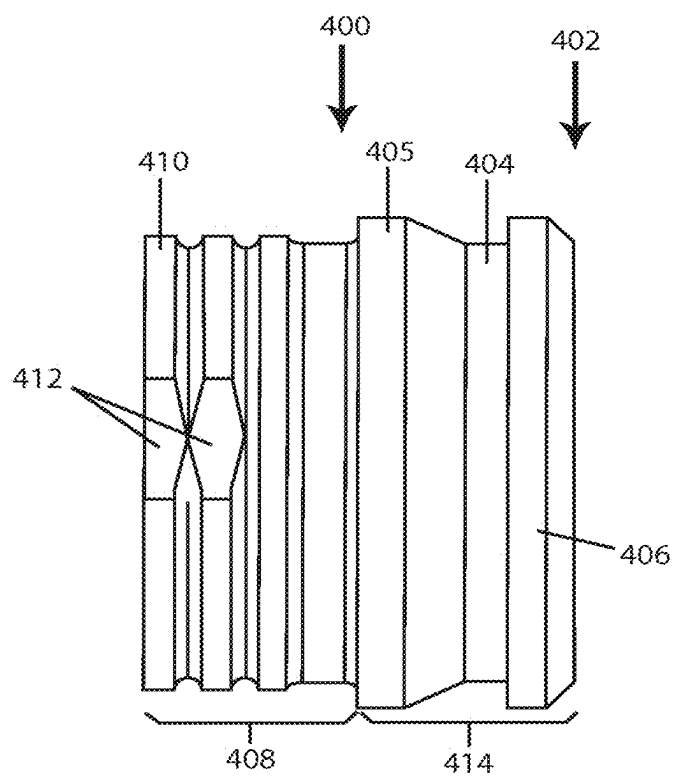


Fig. 9

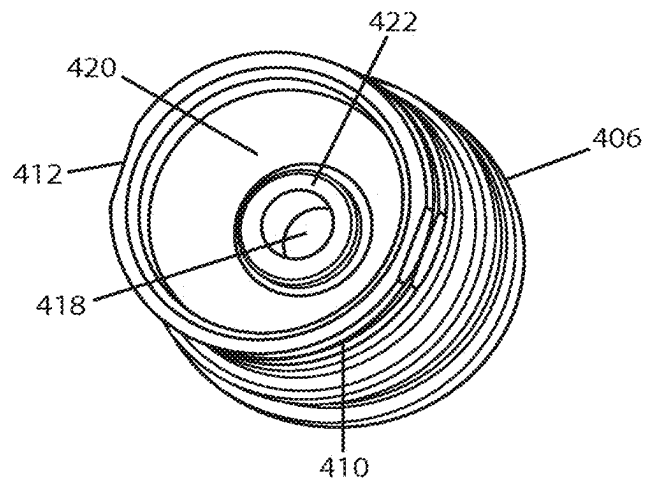


Fig. 10

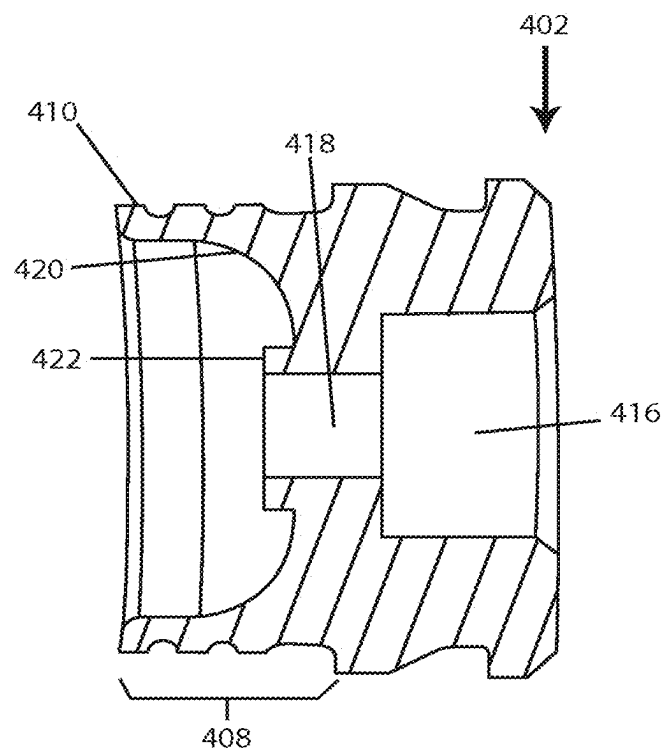


Fig. 11

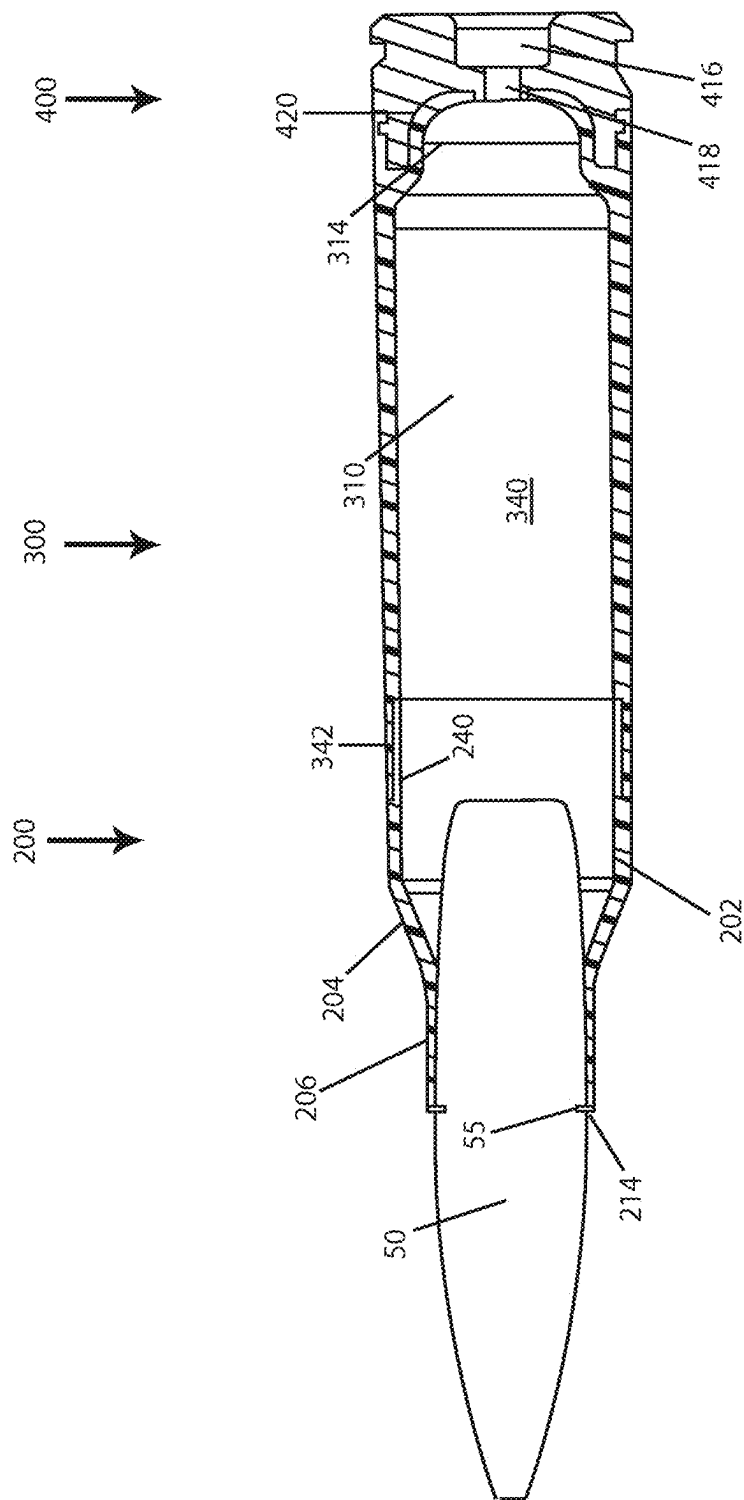


Fig. 12

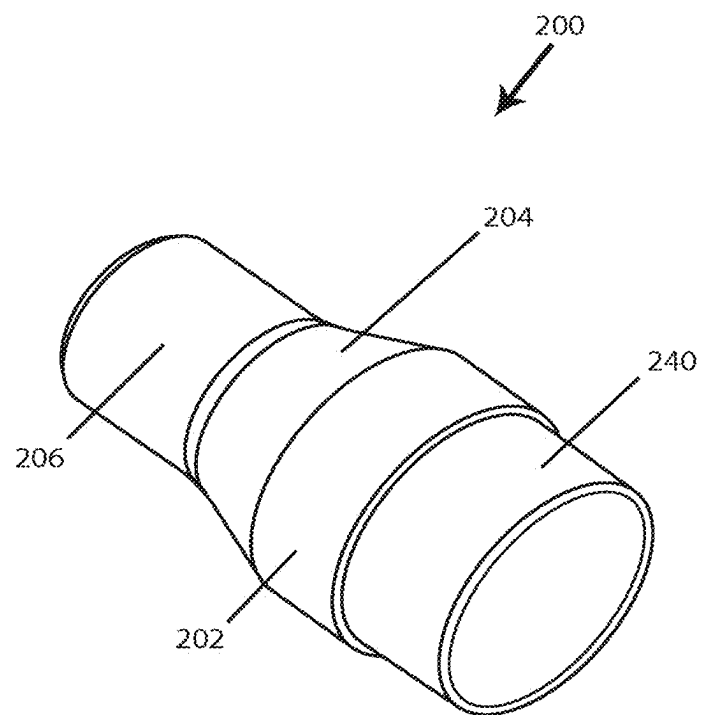


Fig. 13

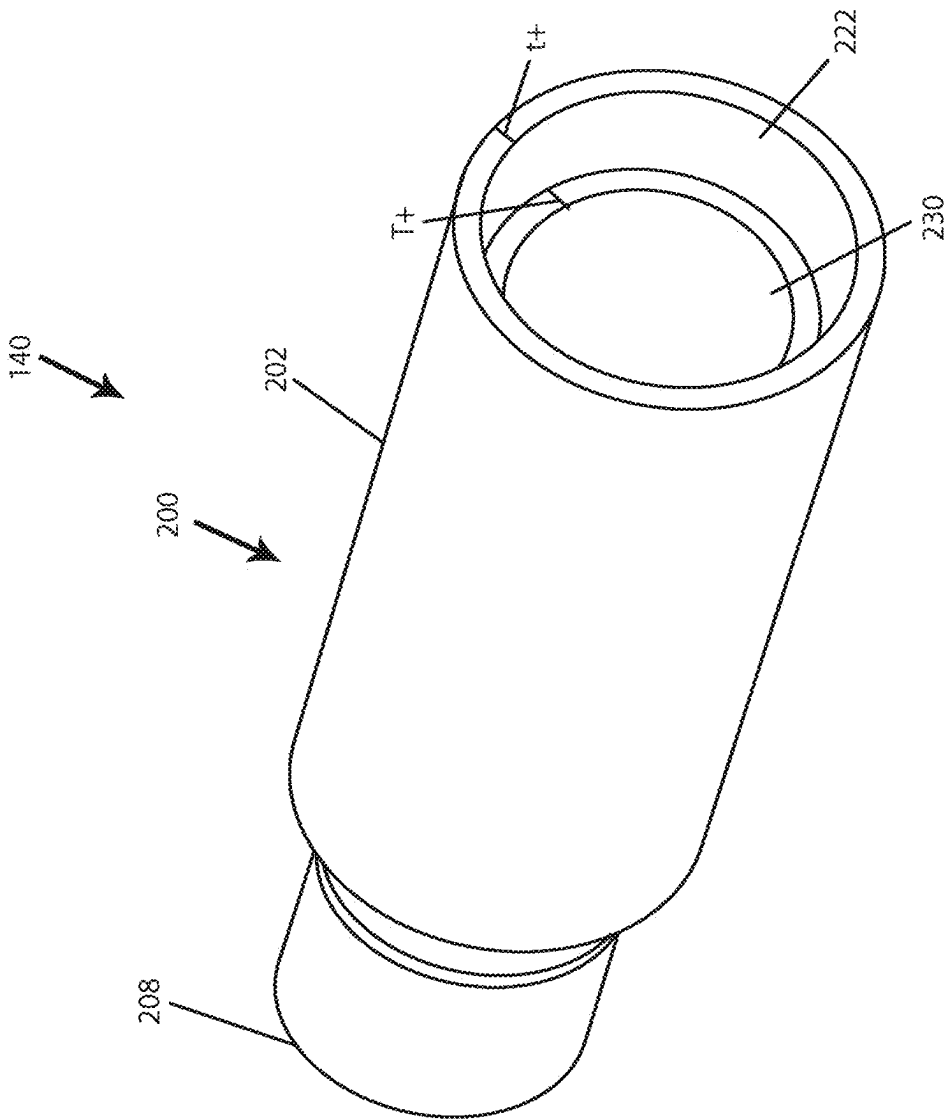


Fig. 14

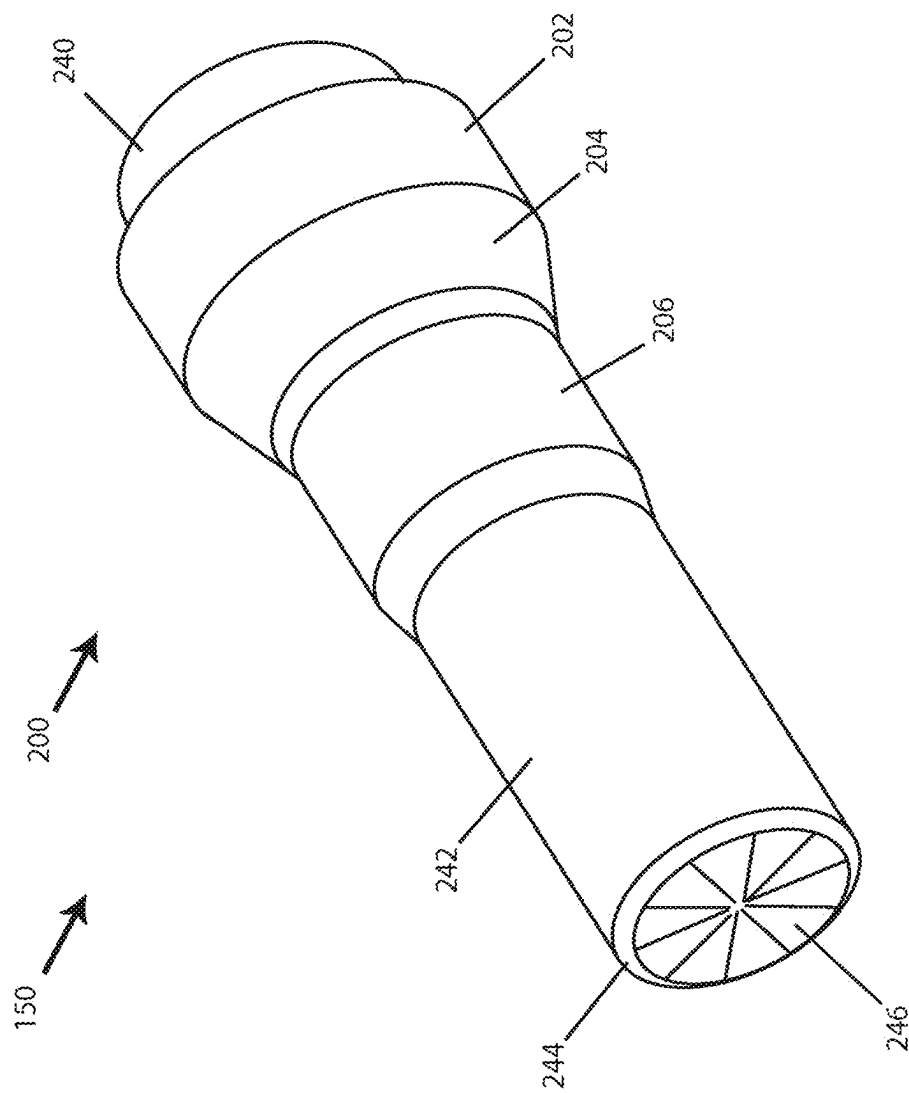


Fig. 15

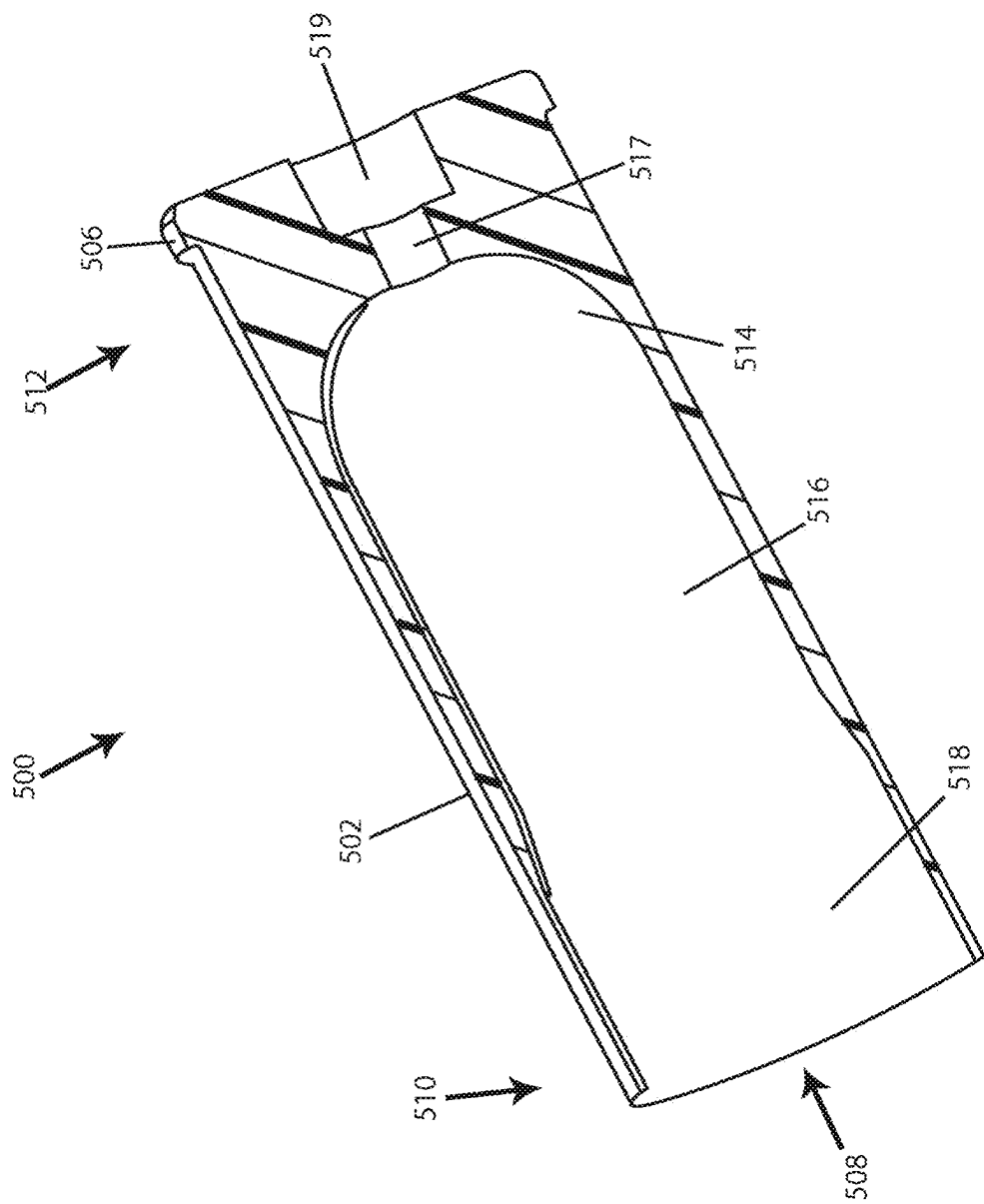


Fig. 16

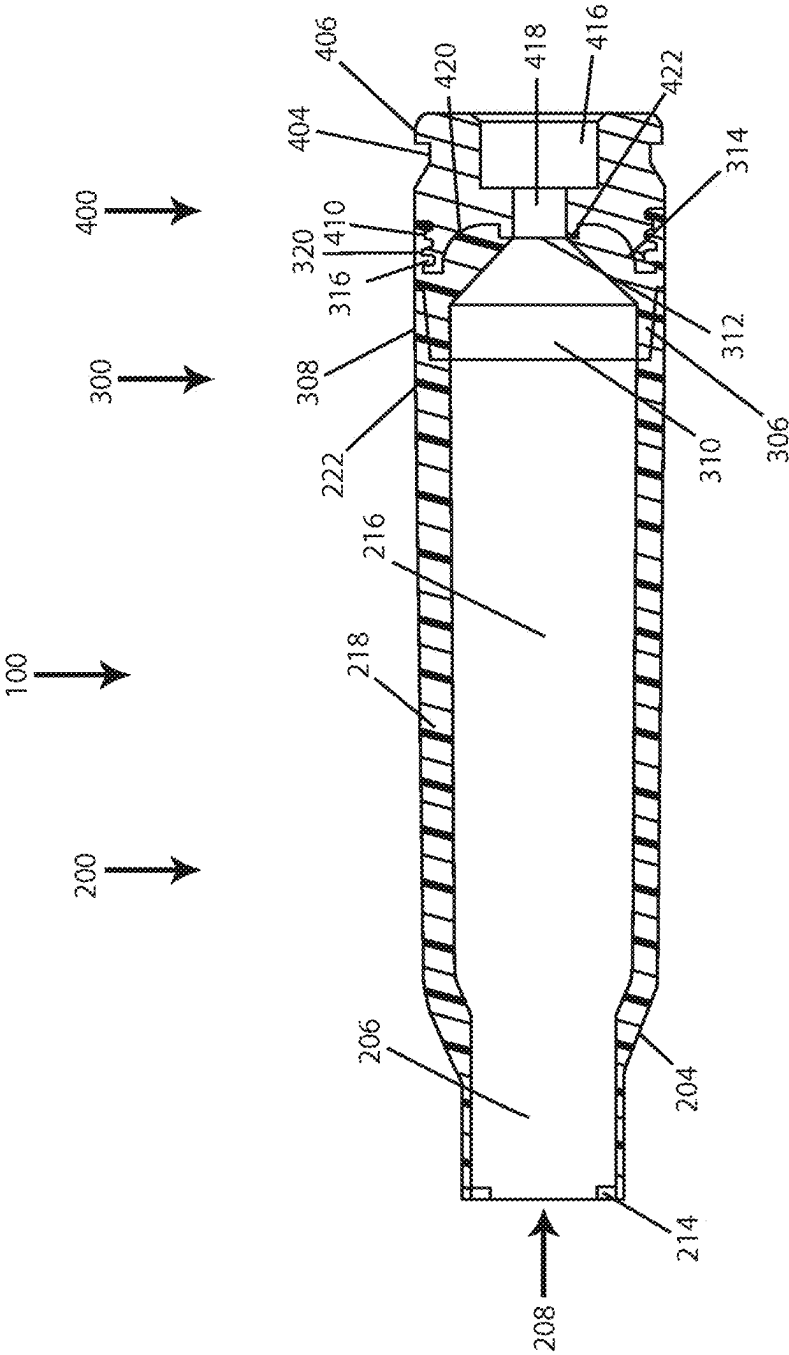
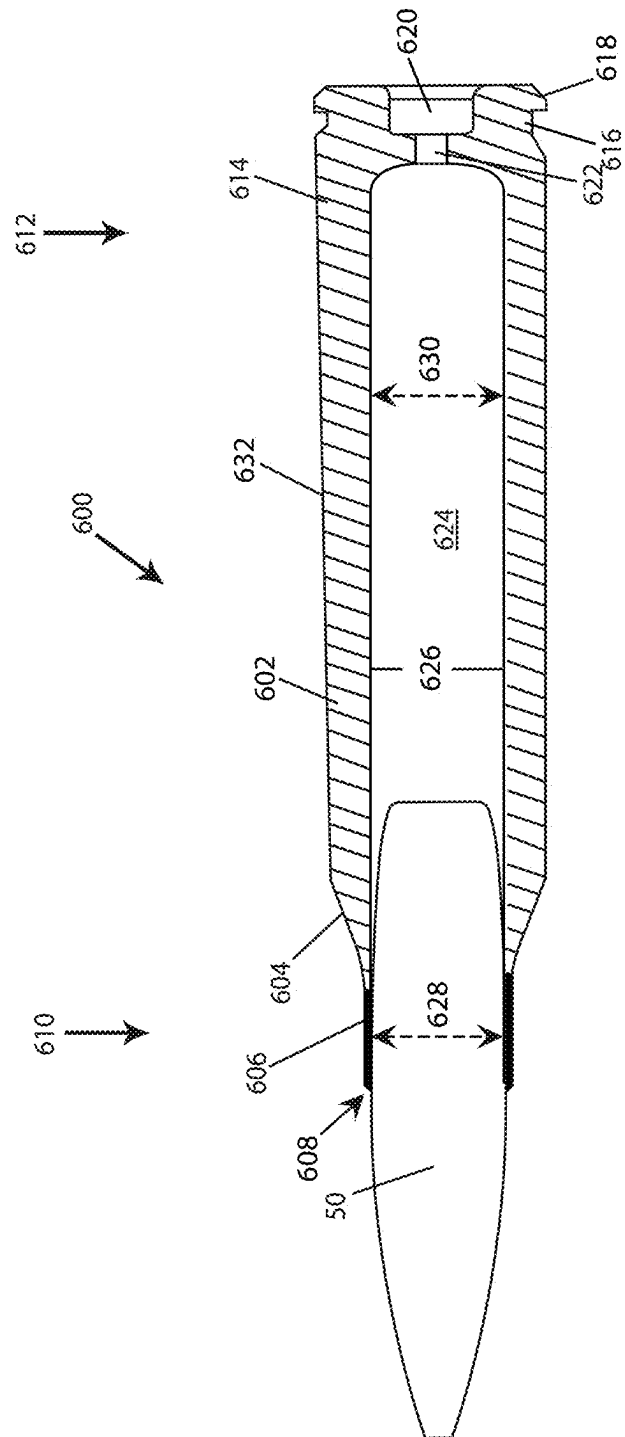


Fig. 17



٨١٥

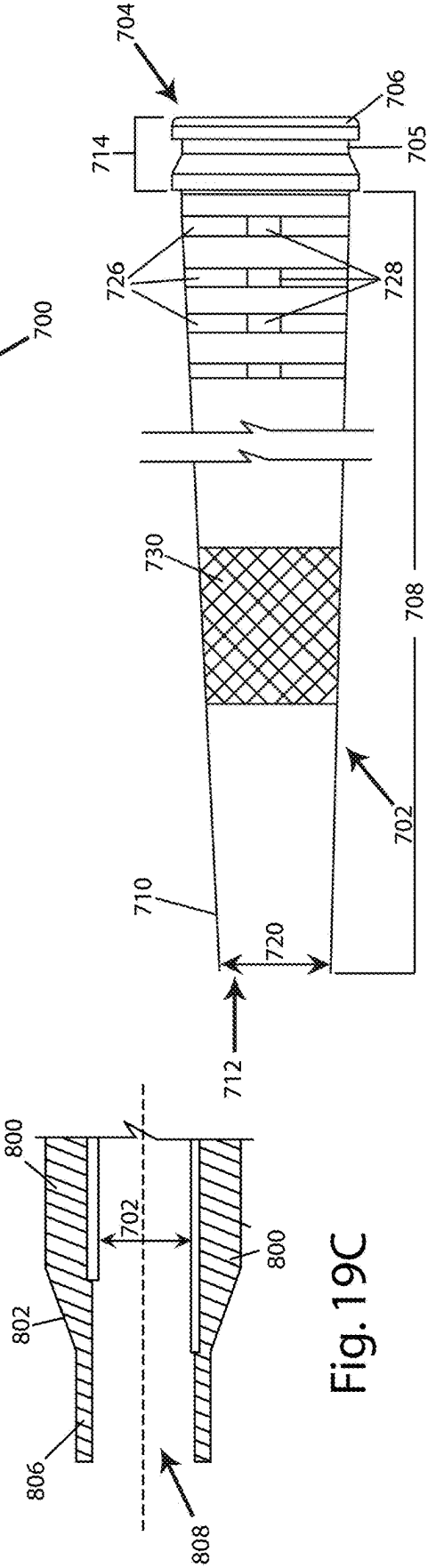
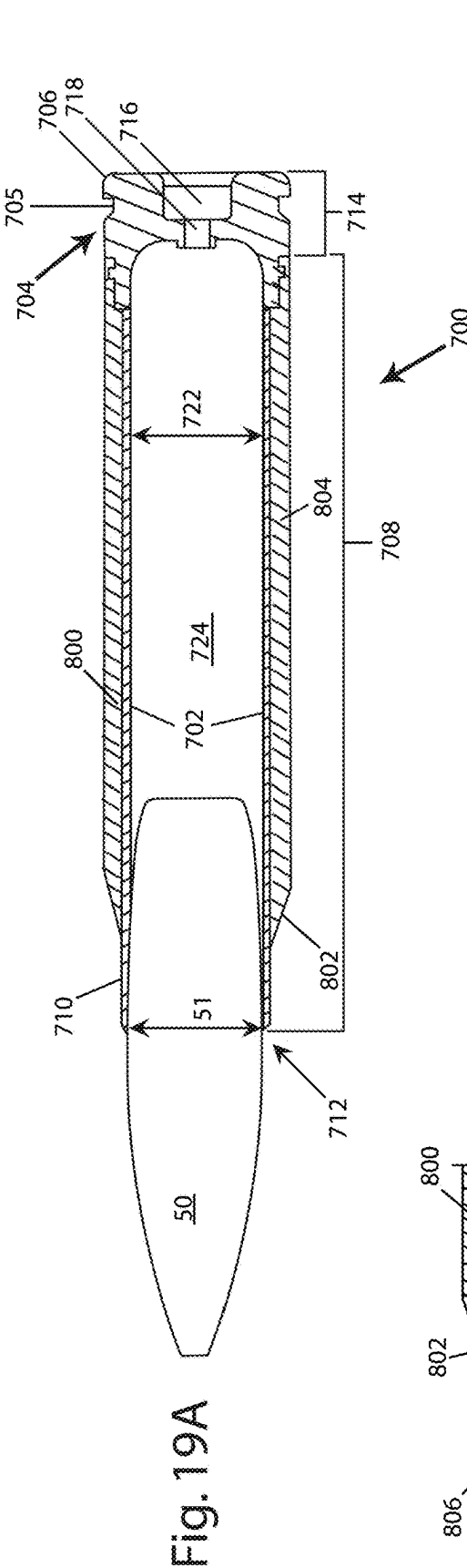


Fig. 19B

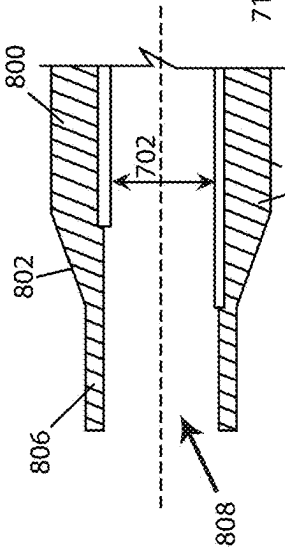


Fig. 19C

Fig. 20A

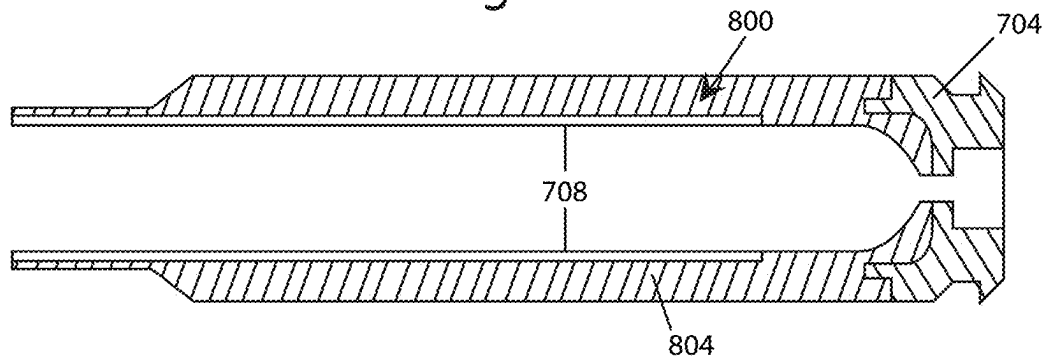


Fig. 20B

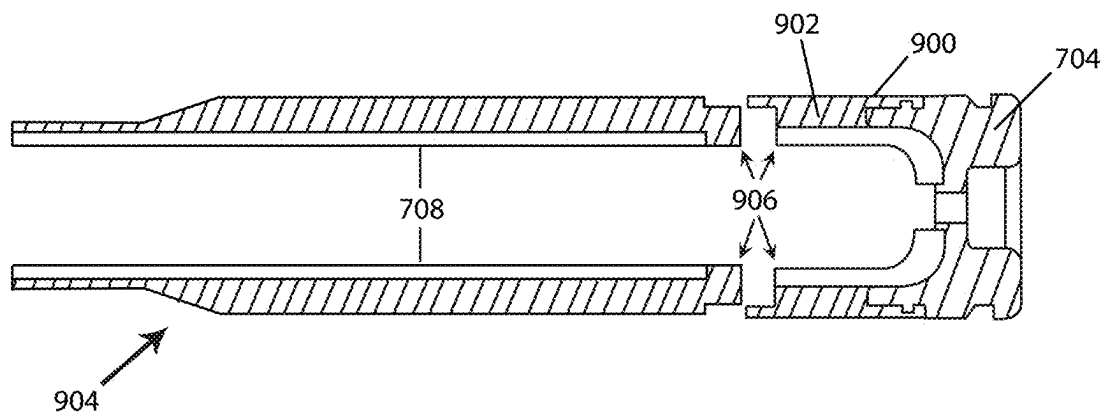
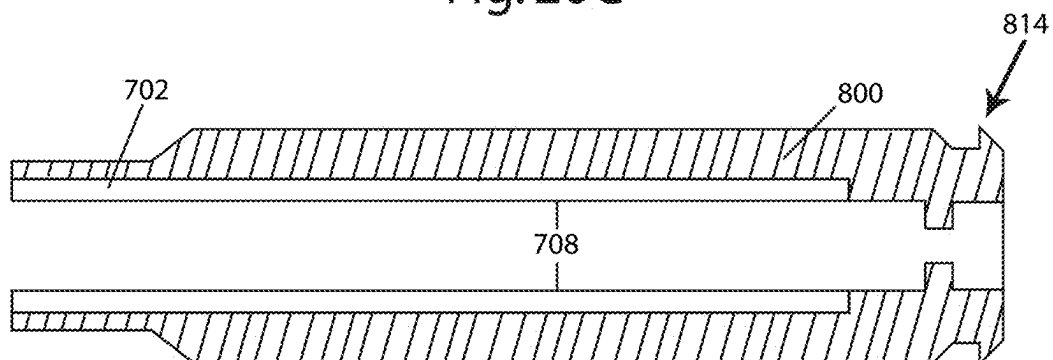


Fig. 20C



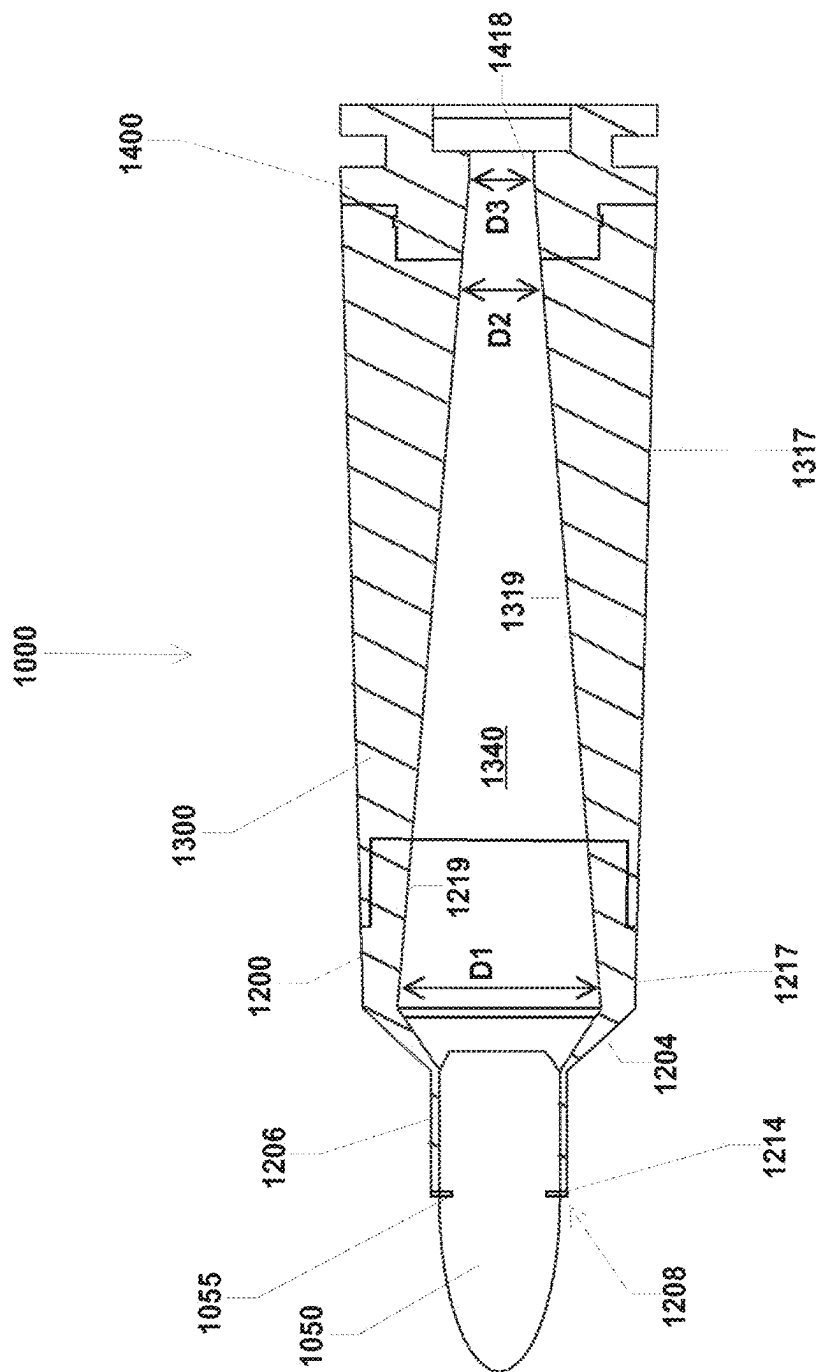


FIG. 21

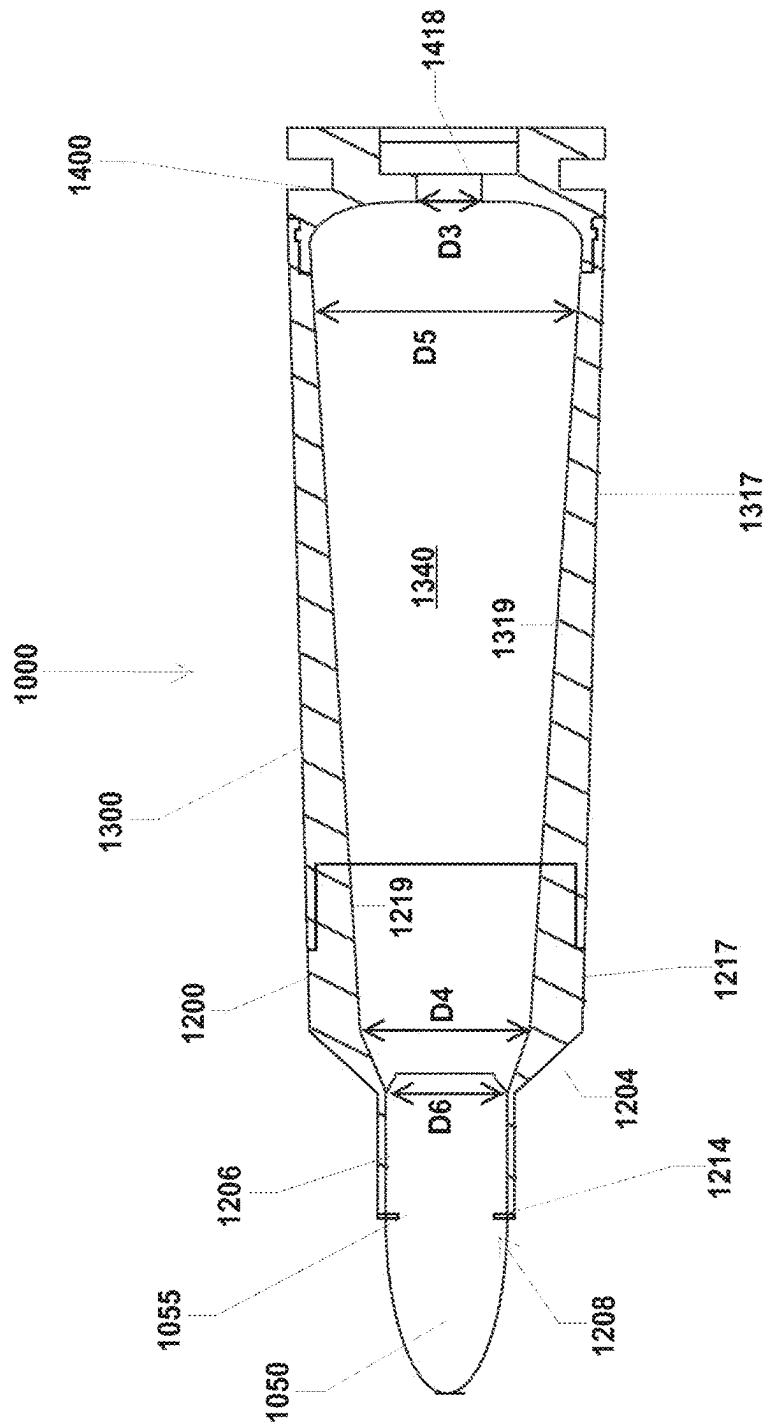


FIG. 22

POLYMER-BASED CARTRIDGE CASING FOR SUBSONIC AMMUNITION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. application Ser. No. 17/833,070 filed Jun. 6, 2022, which in turn is:

A Continuation of U.S. application Ser. No. 17/063,051 filed Oct. 5, 2020, now U.S. Pat. No. 11,353,299, which is a continuation of Ser. No. 16/518,453 filed Jul. 22, 2019, now U.S. Pat. No. 10,794,671, which is a continuation of U.S. application Ser. No. 16/257,262, filed Jan. 25, 2019, now U.S. Pat. No. 10,359,263, which is a Continuation of U.S. application Ser. No. 15/964,911, filed Apr. 27, 2018, now U.S. Pat. No. 10,197,366, which is a Continuation-In Part of U.S. application Ser. No. 15/187,421 filed Jun. 20, 2016, now U.S. Pat. No. 9,995,561, which is a Continuation of U.S. application Ser. No. 14/642,922, filed Mar. 10, 2015, now U.S. Pat. No. 9,372,054, which is a Continuation of U.S. application Ser. No. 14/315,564 filed Jun. 26, 2014, now U.S. Pat. No. 9,003,973, which is a Divisional of U.S. application Ser. No. 13/549,351 filed Jul. 13, 2012, now U.S. Pat. No. 8,763,535, which is Continuation-In-Part of U.S. application Ser. No. 13/350,585, filed Jan. 13, 2012, which claims priority to U.S. Provisional Application No. 61/433,170 filed Jan. 14, 2011.

Further, U.S. application Ser. No. 16/518,453 filed Jul. 22, 2019, now U.S. Pat. No. 10,794,671, which is a Continuation of U.S. application Ser. No. 16/257,262, filed Jan. 25, 2019, now U.S. Pat. No. 10,359,263, which is a Continuation of U.S. application Ser. No. 15/964,911, filed Apr. 27, 2018, now U.S. Pat. No. 10,197,366, which is a Continuation-In-Part of U.S. application Ser. No. 13/828,311, filed Mar. 14, 2013, which is a Continuation-In-Part of U.S. application Ser. No. 13/350,585, filed Jan. 13, 2012 which claims priority to U.S. Provisional Application No. 61/433,170 filed Jan. 14, 2011.

All applications above are incorporated herein by reference.

TECHNICAL FIELD

The present subject matter relates to techniques and equipment to make ammunition articles and, more particularly, to ammunition articles with plastic components such as cartridge casing bodies and bases for at least blank and subsonic ammunition.

BACKGROUND

It is well known in the industry to manufacture bullets and corresponding cartridge cases from either brass or steel. Typically, industry design calls for materials that are strong enough to withstand extreme operating pressures and which can be formed into a cartridge case to hold the bullet, while simultaneously resist rupturing during the firing process.

Conventional ammunition typically includes four basic components, that is, the bullet, the cartridge case holding the bullet therein, a propellant used to push the bullet down the barrel at predetermined velocities, and a primer, which provides the spark needed to ignite the powder which sets the bullet in motion down the barrel.

The cartridge case is typically formed from brass and is configured to hold the bullet therein to create a predetermined resistance, which is known in the industry as bullet

pull. The cartridge case is also designed to contain the propellant media as well as the primer.

However, brass is heavy, expensive, and potentially hazardous. For example, the weight of .50 caliber ammunition is about 60 pounds per box (200 cartridges plus links).

The bullet is configured to fit within an open end or mouth of the cartridge case and conventionally includes a groove (hereinafter referred to as a cannelure) formed in the mid-section of the bullet to accept a crimping action imparted to the metallic cartridge case therein. When the crimped portion of the cartridge case holds the bullet by locking into the cannelure, a bullet pull value is provided representing a predetermined tension at which the cartridge case holds the bullet. The bullet pull value, in effect, assists imparting a regulated pressure and velocity to the bullet when the bullet leaves the cartridge case and travels down the barrel of a gun.

Furthermore, the bullet is typically manufactured from a soft material, such as, for example only, lead, wherein the bullet accepts the mouth of the cartridge being crimped to any portion of the bullet to hold the bullet in place in the cartridge case, even though the cartridge case is crimped to the cannelure of the bullet.

However, one drawback of this design is that the crimped neck does not release from around the bullet evenly when fired. This leads to uncertain performance from round to round. Pressures can build up unevenly and alter the accuracy of the bullet.

The propellant is typically a solid chemical compound in powder form commonly referred to as smokeless powder. Propellants are selected such that when confined within the cartridge case, the propellant burns at a known and predictably rapid rate to produce the desired expanding gases. As discussed above, the expanding gases of the propellant provide the energy force that launches the bullet from the grasp of the cartridge case and propels the bullet down the barrel of the gun at a known and relatively high velocity.

The primer is the smallest of the four basic components used to form conventional ammunition. As discussed above, primers provide the spark needed to ignite the powder that sets the bullet in motion down the barrel. The primer includes a relatively small metal cup containing a priming mixture, foil paper, and relatively small metal post, commonly referred to as an anvil.

When a firing pin of a gun or firearm strikes a casing of the primer, the anvil is crushed to ignite the priming mixture contained in the metal cup of the primer. Typically, the primer mixture is an explosive lead styphnate blended with non-corrosive fuels and oxidizers which burns through a flash hole formed in the rear area of the cartridge case and ignites the propellant stored in the cartridge case. In addition to igniting the propellant, the primer produces an initial pressure to support the burning propellant and seals the rear of the cartridge case to prevent high-pressure gases from escaping rearward. It should be noted that it is well known in the industry to manufacture primers in several different sizes and from different mixtures, each of which affects ignition differently.

The cartridge case, which is typically metallic, acts as a payload delivery vessel and can have several body shapes and head configurations, depending on the caliber of the ammunition. Despite the different body shapes and head configurations, all cartridge cases have a feature used to guide the cartridge case, with a bullet held therein, into the chamber of the gun or firearm.

The primary objective of the cartridge case is to hold the bullet, primer, and propellant therein until the gun is fired.

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Upon firing of the gun, the cartridge case seals the chamber to prevent the hot gases from escaping the chamber in a rearward direction and harming the shooter. The empty cartridge case is extracted manually or with the assistance of gas or recoil from the chamber once the gun is fired.

As shown in FIG. 1A, a bottleneck cartridge case **10** has a body **11** formed with a shoulder **12** that tapers into a neck **13** having a mouth at a first end. A primer holding chamber **15** is formed at a second end of the body opposite the first end. A divider **16** separates a main cartridge case holding chamber **17**, which contains a propellant, from the primer holding chamber **15**, which communicate with each other via a flash hole channel **18** formed in the web area **16**. An exterior circumferential region of the rear end of the cartridge case includes an extraction groove **19a** and a rim **19b**.

Prior art patents in this area include U.S. Pat. No. 4,147,107 to Ringdal, U.S. Pat. No. 6,845,716 to Hussein et al., U.S. Pat. No. 7,213,519 to Wiley et al., and U.S. Pat. No. 7,610,858 to Chung. The four patents are directed to an ammunition cartridge suitable for rifles or guns and including a cartridge case made of at least a plastics material. However, each have their own drawbacks.

Further, the use of brass cartridges for blank or subsonic ammunition can be problematic. To reduce the velocity of the bullet exiting the cartridge, typically less propellant is used in comparison to when the bullet is traveling at its top velocity. However, the same size cartridge needs to be used so the bullet can be fired from a standard firearm. An empty space is left inside a blank or subsonic cartridge where the propellant would normally reside. To compensate, wadding (typically cotton) can be packed into the space normally filled by the propellant. This wadding can cause problems with the use of the round, including jamming the firearm and fouling silencers and/or suppressors attached to the firearm.

Other inventions attempting to address some of the above issues include U.S. Pat. No. 6,283,035 to Olsen, which places an expanding insert into a brass cartridge, and U.S. Patent Application Publication No. 2003/0019385 to Lea-Sure which uses a heavier than standard bullet with a reduced capacity cartridge.

Hence, a need exists for a polymer casing that can perform as well as or better than the brass alternative. A further improvement is polymer casings that are capable of production in a more conventional and cost-effective manner, i.e. by using standard loading presses. Additionally, the cartridge can provide increased performance for blank and subsonic rounds.

SUMMARY

The invention includes examples of a high strength polymer-based cartridge for subsonic ammunition with a first end having a mouth, a projectile disposed in the mouth, a shoulder disposed below the mouth forming a bottleneck cartridge; and at least a wall, molded from a polymer, between the first end and a second end opposite the first end. Further, included is an insert joined to the second end, having an extraction rim and a groove both disposed at one end of the insert; and a primer pocket in fluid communication with a flash hole, the flash hole in fluid communication with a propellant chamber. A sleeve section is also included and the sleeve section and the wall form the propellant chamber and have a thickness at least 1.25 times greater than a standard thickness of a wall of a standard cartridge. The propellant chamber between the mouth and the insert is unobstructed and comprises a powder load having a load density greater than 40%.

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Examples of the high strength polymer-based cartridge have the sleeve section with a first inner wall having a first diameter; and a second inner wall having a second diameter. The first inner wall extends from the shoulder to the second inner wall and the second inner wall extends from the upper inner wall to the insert. Further, the first diameter does not equal the second diameter.

In other examples, the sleeve section further includes a first inner wall having a first slope, and a second inner wall having a second slope. The first slope can extend between the shoulder and the second inner wall while the second slope can extend between the first inner wall and the insert. The first slope may not equal the second slope,

In examples, the propellant chamber permits only enough propellant to propel the projectile engaged in the cartridge casing at subsonic speeds.

As a result, a light weight, high strength cartridge case can be formed using standard brass cartridge loading equipment. As noted below, the present invention can be adapted to any type of cartridge, caliber, powder load, or primer. Calibers can range at least between 0.22 and 30 mm and accept any type of bullet that can be loaded in a typical brass cartridge.

Further advantages can be gained in both blank and subsonic ammunition due to the removal of wadding and the shrinking of the volume of powder based on a reduced volume in the cartridge.

The polymer used can be of any known polymer and additives, but the present invention uses a nylon polymer with glass fibers. Further, the portion of the cartridge that engages the extractor of the firearm can be made from heat strengthened steel for normal loads and can be a continuous molded polymer piece of the lower component for either subsonic or blank ammunition.

Additional advantages and novel features will be set forth in part in the description which follows, and in part will become apparent to those skilled in the art upon examination of the following and the accompanying drawings or may be learned by production or operation of the examples. The advantages of the present teachings may be realized and attained by practice or use of various aspects of the methodologies, instrumentalities and combinations set forth in the detailed examples discussed below.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements.

FIG. 1A is a cross sectional view of a conventional bottleneck cartridge case;

FIG. 1B is a side view of a conventional bullet;

FIG. 2 is a side perspective view of the outside of cartridge case of the present invention;

FIG. 3 is a longitudinal cross-section of the upper component of the cartridge;

FIG. 4 is a bottom, side, perspective, radial cross-section of the upper and lower components of the cartridge;

FIG. 5 is an end view of the upper component without the lower component and insert;

FIG. 6 is a side view of the lower component without the upper component and insert;

FIG. 7 is a bottom front perspective view of the lower component of FIG. 6;

FIG. 8 is a longitudinal cross-section view of the lower component of FIG. 6;

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FIG. 9 is a side view of the insert without the upper and lower components;

FIG. 10 is a bottom front perspective view of the insert of FIG. 8;

FIG. 11 is a longitudinal cross-section view of the insert of FIG. 8;

FIG. 12 is a longitudinal cross-section view of an example of a cartridge case;

FIG. 13 is a top, side, perspective view of the upper component of the example;

FIG. 14 is a top, side perspective view of an example of an upper component of a subsonic cartridge;

FIG. 15 is a top, side perspective view of an upper component for a blank cartridge;

FIG. 16 is a longitudinal cross-section view of an example of a straight wall cartridge case;

FIG. 17 is a longitudinal cross-section view of the cartridge case of FIG. 2;

FIG. 18 is a longitudinal cross-section view of an example of a one-piece blank or subsonic cartridge case;

FIG. 19A is a longitudinal cross-section view of an example of a metallic sleeve with a polymer sheath for a blank or subsonic cartridge case;

FIG. 19B is a side view of an example of the metallic sleeve of FIG. 19A;

FIG. 19C is a partial split longitudinal cross-section view of an example of a polymer neck with the metallic sleeve;

FIG. 20A is a longitudinal cross-section view of an example of a two-part metallic sleeve with a one-piece blank or subsonic cartridge case;

FIG. 20B is a longitudinal cross-section view of an example of a two-part metallic sleeve with a two-piece blank or subsonic cartridge case;

FIG. 20C is a longitudinal cross-section view of an example of a one-part metallic sleeve with a one-piece blank or subsonic cartridge case;

FIG. 21 is a longitudinal cross-section view of an example of a tapered wall cartridge case; and

FIG. 22 is a longitudinal cross-section view of another example of a tapered wall cartridge case.

DETAILED DESCRIPTION

In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent to those skilled in the art that the present teachings may be practiced without such details. In other instances, well known methods, procedures, components, and/or circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present teachings.

The present invention provides a cartridge case body strong enough to withstand gas pressures that equal or surpass the strength of brass cartridge cases under certain conditions, e.g. for both storage and handling.

Reference now is made in detail to the examples illustrated in the accompanying drawings and discussed below. FIG. 2 illustrates an example of a cartridge case 100. The cartridge case 100 includes an upper component 200, a lower component 300, and an insert 400. In this example, the upper component 200 and the lower component 300 are made of a polymer, while insert 400 is made from a metal, an alloy of metals, or an alloy of a metal and a non-metal. Regardless of materials, the outer dimensions of the cartridge case 100 are within the acceptable tolerances for whatever caliber firearm it will be loaded into.

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The polymer used is lighter than brass. A glass-filled high impact polymer can be used where the glass content is between 0%-50%, preferably between 5% and 50%. In another example the glass content can be 10%. An example of a high impact polymer without the glass content is BASF's Capron® BU50I. The insert 400 can be made of steel, and, in an example, heat treated carbon steel, 4140. The 4140 steel is further heat treated to a Rockwell "C" scale ("RC") hardness of about 20 to about 50. However, any carbon steel with similar properties, other metals, metal alloys or metal/non-metal alloys can be used to form the insert. Heat treating a lower cost steel alloy to improve its strength is a point of distinction from the prior art, which have typically opted for more expensive alloys to deal with the strength and ductility needed for a cartridge casing application.

Further to the above, as noted for the insert, any metal, metal alloy, or non-metal alloys, ranging from the common place (e.g. brass) to the exotic (e.g. ceramics), can be used to form the insert. The main requirement is to withstand both the explosive and subsequent extractive forces subjected to the insert. The ability to form the insert easily and inexpensively are of a separate consideration. The same holds true for the polymer, it can be of any type or quality as long as it meets the requirements of the specific example noted below.

In an example, the combination of the upper component 200 and the lower component 300 are made of 10% glass-filled high impact polymer combined with the insert 400 made of heat treated 4140 steel results in a cartridge that is approximately 50% lighter than a brass formed counterpart. This weight savings in the unloaded cartridge produces a loaded cartridge of between 25%-30% lighter than the loaded brass cartridge depending on the load used, i.e. which bullet, how much powder, and type of powder used.

The upper component 200 includes a body 202 which transitions into a shoulder 204 that tapers into a neck 206 having a mouth 208 at a first end 210. The upper component 200 joins the lower component 300 at an opposite, second end 212. The lower component 300 joins the upper component 200 at a lower component first end 302 (see FIG. 6). The upper 200 and lower 300 components are adhered by an ultraviolet (UV) light or heat cured resin, a spin weld, a laser weld or an ultrasonic weld.

At a second end 304 of the lower component 300, the lower component is joined to the insert 400. In one example, the upper component 200 and the lower component 300 are molded in separate molds. When the lower component 300 is molded, it is molded over the insert 400. This is a partial molding over, since the lower component 300 does not completely cover the insert 400.

A back end 402 of the insert 400 is also the rear end of the casing 100. The insert 400 is formed with an extraction groove 404 and a rim 406. The groove 404 and rim 406 are dimensioned to the specific size as dictated by the caliber of the ammunition. The insert 400 can be formed by turning down bar stock to the specific dimensions or can be cold formed.

Turning now to FIG. 3, a cross-section of the upper component 200 is illustrated. Because of the nature of the polymer, and the design of the neck 206 and mouth 208, the neck 206 expands uniformly under the gas pressures formed during firing. This concentric expansion provides a smoother release of the projectile into the barrel of the firearm. The smoother release allows for a more stable flight of the projectile, providing greater accuracy and distance with the same amount of powder.

Moving toward the second end **212** of the upper component **200**, as the neck **206** transitions into the shoulder **204**, a sleeve **230** begins. The sleeve **230**, in this example, extends approximately to the second end **212**. The sleeve **230** can be an additional thickness to a wall **218** as is normally required for a standard cartridge, or a separately manufactured and adhered to the wall **218**. The sleeve **230** provides additional strength relative to the wall **218** of the body **202** alone. This strengthening, which is in the lateral direction, reduces bending of the upper component **200** of the cartridge case **100**. The sleeve **230** helps to keep the cartridge **100** as concentric as possible, and as noted above, concentricity is a key to accuracy.

The case wall **218** can have a thickness T , and the sleeve **230** can have a thickness $T+$, as illustrated in FIG. 4. Thus, the total thickness of the cartridge at the point where there is the wall **218** and sleeve **230** is the sum of T and $T+$.

The upper portion **220** of the sleeve **230** can begin in or near the neck **206** and extend over the shoulder **204**. In one example, the upper portion **220** of the sleeve **230** ends against a bullet **50** (see FIG. 1B) providing additional material, and thus strength, to help retain and align the bullet **50**. This thickened upper portion **220** can act like an extension of the neck **206** farther down into the shoulder. The upper portion **220** is an advantage over a brass cartridge, since brass cannot be formed in this way. Thus, the upper portion **220** can act to sit and secure the bullet in the same place in the cartridge every time.

The sleeve **230**, in the illustrated example of FIGS. 3, 4 and 5, extends almost the entire length of the body **202**. The sleeve **230** stops at an overlap portion **222** of the upper component **200**. The overlap portion **222** is the portion of the upper component **200** that engages the lower component **300**. The overlap portion **222** has a thinner wall thickness t , or a second thickness, at the second end **212** than the thickness T of the wall **218** (or T and $T+$) before the overlap portion **222**. The second thickness t tapers toward the outside of the upper component **200** so an outer diameter **224** of the wall **218** remains constant while an inner diameter **226** of the wall **218** increases. This allows certain examples of cartridge **100** to maintain a constant outer diameter from below the shoulder **204** to the insert **400**. The bottom end **228** of the sleeve **230** is approximately squared off to provide a square shoulder to keep the upper **200** and lower **300** components concentric during assembly.

FIGS. 6-8 illustrate that the lower component **300** has a tapered portion **306** starting at the lower component first end **302** and ending at a collar **308**. The slope of the tapered portion **306** approximately matches the slope of the overlap portion **222** so the two can slide over each other to engage the upper **200** and lower **300** components. The tapered portion **306** ends in a flat seat **307**. The seat **307** can have a thickness T_s which is about equal to the thickness of the wall and/or sleeve. This allows the bottom end **228** of the sleeve to sit on the seat **307** when the upper **200** and lower **300** components engage. This prevents the bottom end **228** of the sleeve **230** from being exposed. This could allow the gases to exert pressure on the bottom end **228** that can separate the upper **200** from the lower **300** component.

A width of the collar **308** matches the second thickness t , so that the outer diameter of the cartridge **100** remains constant past the transition point between the upper **200** and lower **300** components. Further, a thickness of the tapered portion **306** is such that at any point the sum of it with the thickness of the overlap portion **222** is approximately equal to the thickness T of the wall **218** or the thicknesses of the wall **218** and sleeve **230** (T and $T+$). As noted above, the

tapered portion **306** and the overlap portion **222** are bonded together to join the upper **200** and lower **300** components.

An inner wall **310** of the lower component **300** can be formed straight. In the illustrated example in FIG. 8, the inner wall **310** forms a bowl shape with a hole **312** at the bottom. The hole **312** is formed as a function of the interface between the lower component **300** and the insert **400**, and its formation is discussed below. As the inner wall **310** slopes inward to form the bowl shape, it forks and forms an inner bowl **314** and an outer sheath **316**. The gap **318** that is formed between the inner bowl **314** and the outer sheath **316** is the space where a portion of the insert **400** engages the lower component **300**. As noted above, in one example, the lower component **300** is molded over a portion of the insert **400** to join the two parts.

Turning now to an example of the insert **400**, as illustrated in FIG. 9, it includes an overmolded area **408**, where the outer sheath **316** engages the insert **400** in the gap **318**. The overmolded area **408** has one or more ridges **410**. The ridges **410** allow the polymer from the outer sheath **316**, during molding, to form bands **320** (see, FIG. 8) in the gap **318**. The combination of the ridges **410** and bands **320** aid in resisting separation between the insert **400** and the lower component **300**. The resistance is most important during the extraction of the cartridge from the firearm by an extractor (not illustrated).

The overmolded area **408** also includes one or more keys **412**. The keys **412** are flat surfaces on the ridges **410**. These keys **412** prevent the insert **400** and the lower portion **300** from rotating in relation to one another, i.e. the insert **400** twisting around in the lower portion **300**.

Below the overmolded area **408**, toward the back end **402**, is a self-reinforced area **414**. This portion extends to the back end **402** of the insert **400** and includes the extraction groove **404** and rim **406**. The self-reinforced area **414** must, solely by the strength of its materials, withstand the forces exerted by the pressures generated by the gasses when firing the bullet and the forces generated by the extractor. In the present example, the self-reinforced area **414** withstands these forces because it is made of a heat treated metal or a metal/non-metal alloy.

FIGS. 10 and 11 illustrate an example of the inside of the insert **400**. Open along a portion of the back end **402** and continuing partially toward the overmolded area **408** is a primer pocket **416**. The primer pocket **416** is dimensioned according to the standards for caliber of the cartridge case and intended use. A primer (not illustrated) is seated in the primer pocket **416**, and, as described above, when stricken causes an explosive force that ignites the powder (not illustrated) present in the upper **200** and lower **300** components.

Forward of the primer pocket **416** is a flash hole **418**. Again, the flash hole **418** is dimensioned according to the standards for the caliber of the cartridge case and intended use. The flash hole **418** allows the explosive force of the primer, seated in the primer pocket **416**, to communicate with the upper **200** and lower **300** components.

Forward of the primer pocket **416** and inside the overmolded area **408** is basin **420**. The basin **420** is adjacent to and outside of the inner bowl **314** of the lower component **300**. The basin **420** is bowl shaped, wherein the walls curve inwards toward the bottom. The bottom of the basin **420** is interrupted by a ring **422**. The ring **422** surrounds the flash hole **418** and extends into the basin **420**. It is the presence of the ring **422** that forms the hole **312** in the inner bowl **314** of the lower component **300**.

In another example of a cartridge case **120**, the sizes of the upper **200** and lower **300** components can be altered. FIG. **12** illustrates a “small upper” embodiment with a bullet **50** in the mouth **208** of the cartridge **120**. The features of the upper **200** and lower **300** component are almost identical to the example discussed above, and the insert **400** can be identical. FIG. **12** also illustrates the engagement between a lip **214** and the cannellure **55**. The lip **214** is a section of the neck **206** approximate to the mouth **208** that has a thicker cross section or, said differently, a portion having a smaller inner diameter than the remainder of the neck **206**. In this example, the lip **214** is square or rectangular shaped, no angles or curves in the longitudinal direction. Note, in other examples, the upper component **200** is not formed with a lip **214**. When present, the lip **214** engages the cannellure **55** formed along an outer circumferential surface of the bullet **50** when it is fitted into the mouth **208** of the cartridge casing **100**.

FIG. **13** shows that the neck **206** and the shoulder **204** are formed similar, but in this example, the body **202** is much shorter. Further, instead of an overlap portion **222**, there is an underskirt portion **240** that starts very close to the shoulder **204**. The underskirt portion **240** tapers to the inside of the cartridge when it engages the lower component **300**.

The lower component **300** in this further example, is now much longer and comprises most of the propellant chamber **340**. The tapered portion is now replaced with an outer tapered portion **342**. The outer tapered portion **342** slides over the underskirt portion **240** so the two can be joined together as noted above. The thickness of the underskirt portion **240** and the outer tapered portion **342** is approximate to the wall thickness or wall thickness and sleeve thickness.

The inner wall **310** is now substantially longer, can include a sleeve, but still ends in the inner bowl **314**. The engagement between the second end **304** of the lower component **300** and the insert **400** remains the same. Note that either the “small upper” or “long upper” can be used to form blank or subsonic ammunition. The walls are made thicker with the sleeve, shrinking the size of the propellant chamber **340**. Less powder can be used, but the powder is packed similarly as tight as it is for a live round because of the smaller chamber **340**. This can prevent the Secondary Explosive Effect (SEE) (below). A thick wall design for a subsonic cartridge **140** is illustrated in FIG. **14**.

Illustrated is a large upper component **200** having a thicker overlap **222** portion, with a thickness $t+$ and an integral thickening of the wall, and/or a sleeve **230** with a thickness $T+$, as disclosed above. The total thickness of the wall **218** can be the sum of $T+$ and $t+$. The sleeve **230** can run the length of the upper component **200** from the mouth **208** to the start of the overlap portion **222**. The lower component **300** of a subsonic cartridge **140** can be thickened as well. The subsonic cartridge **140** can be made with the insert **400**, or the lower component **300** can be molded in one piece from polymer with the features of the insert **400**. For example, the flash hole **418**, primer pocket **416**, groove **404** and rim **406**. Alternately, the insert can also be high-strength polymer instead of the metal alloys discussed above. In this example, the lower component and the insert can be formed as one piece, and the upper component **200** can be placed on top.

As illustrated in FIG. **15**, for a blank cartridge **150**, the upper component **200** can be made differently. For the blank cartridge **150**, an extension **242** can be molded to extend from the neck **206**. The extension **242** has a star-shaped cap **244** to seal off the cartridge. The cap **244** is formed partially of radially spaced fingers **246** that deform outwards during

firing. Thus, the mouth **208** is molded partially shut to contain a majority of the pressures and expand open and outwards. The fingers **246** are designed, in one example, to be bend elastically and are not frangible. The object is to contain the majority of the pressures and expel anything that can act as a projectile out the barrel of the firearm.

When the blank cartridge **150** is formed with the “small upper” component **200** with the cap **244**. The lower component **300** can be filled with the powder and the small upper component can act as a cap to the cartridge, sealing in the powder.

Note that the above examples illustrate a bottleneck cartridge. Many of the features above can be used with any cartridge style, including straight wall cartridges used in pistols. FIG. **16** illustrates an example of a straight wall cartridge **500**. The straight wall cartridge **500** is a one-piece design of all polymer. The cartridge **500** has a body **502** and a mouth **508** at a first end **510**. The walls **518** of the cartridge casing can also have a sleeve **530** along a majority of its length.

The sleeve **230**, **530** is dimensioned and shaped pursuant to the requirements of each cartridge based on blank or subsonic and the particular caliber. To that end, the sleeve **530** begins set back from the first end **510** based on the depth the rear of the bullet sits in the cartridge. Further, in this example, as the walls transition into a lower bowl **514**, the sleeve **530** may extend into the bowl. This aids in the strength of a back end **512** of the cartridge **500**, since this example lacks a hardened metal insert.

The lower bowl **514** curves downward toward a flash hole **517** which then opens to a primer pocket **519**. Both are similar to the features described above. Further, the back end is molded to form a rim **506**.

Turning now to an example of a fully formed cartridge case **100**, FIG. **17** illustrates a cross-section of all three elements engaged together to illustrate how they interface with each other. The specific outer dimensions of the three elements and certain inner dimensions (e.g. mouth **208**, lip **214**, flash hole **418**, and primer pocket **416**) are dictated by the caliber and type of the firearm and type of ammunition. The cartridge casing **100** of the present invention is designed to be used for any and all types of firearms and calibers, including pistols, rifles, manual, semi-automatic, and automatic firearms.

An exemplary construction of the upper component **200** also aids in withstanding the pressures generated. As noted above, the sleeve **230** increases the strength of the wall **218** of the upper component **200**. In the present example, the upper component **200** accounts for anywhere from 70% to 90% of the length of the cartridge casing **100**.

The polymer construction of the cartridge case also provides a feature of reduced friction between the cartridge and chamber of the firearm. Reduced friction leads to reduced wear on the chamber, further extending its service life.

Turning now to FIG. **18**, an example of a one-piece subsonic cartridge casing **600** is illustrated. In this example, the entire cartridge casing **600** is polymer. The subsonic cartridge casing **600** includes a body **602** which, at a first end **610** transitions into a shoulder **604** that tapers on the outside into a neck **606** having a mouth **608**. The bullet **50** can be inserted into the mouth **608** of the subsonic cartridge casing **600**.

Opposite the first end **610** is second end **612**. A back end **614** is the rear of the second end **612** of the subsonic cartridge casing **600**. The back end **614** is formed with an extraction groove **616** and a rim **618**. The groove **616** and rim **618** are dimensioned to the specific size as dictated by

the caliber of the ammunition. Also, included in the back end **614** is a primer pocket **620**. The primer pocket **620** is dimensioned according to the standards for caliber of the cartridge case and intended use. Forward of the primer pocket **620** is a flash hole **622**. Again, the flash hole **622** can be dimensioned according to the standards for the caliber of the cartridge case and intended use. The flash hole **622** allows the explosive force of the primer, seated in the primer pocket **620**, to communicate with a propellant chamber **624**.

In this example, the propellant chamber **624** is formed from the inner wall **626** of the body **602**. The inner wall **626** can be straight from the mouth **608** to the back end **214**. Thus, a first diameter **628** of the inside of the mouth **608** is approximately equal to a second diameter **630** of the propellant chamber **624**. Alternately, or in addition to, the first diameter **628** can be a diameter of the inside of the neck **606**.

An outside wall **632** of the body **602** is shaped and dimensioned according to the standards for the caliber of the cartridge case and intended use. This includes the length of the neck, the angle of the shoulders, and length of the total body. A straight inner wall **626** acts to thicken the walls of the cartridge **600**, providing the benefits as described above. The thickened walls act to reduce the size of the propellant chamber **624**, allowing less powder to be used. In certain examples this can generate lower pressures on ignition and expel the bullet **50** at subsonic speeds.

The straight inside wall **626** example makes for ease of molding. A single "pin" or mandrel can be set to mold a constant diameter from mouth/neck **608**, **606** to back end **614**. The back end **614** can also be made of polymer. Since examples of the cartridge **600** are designed to generate lower pressures, certain calibers or designs do not require the insert **400**, as described above.

In other examples, the subsonic cartridge casing **600** can be either formed from 2 or 3 parts. In one example, the back end **614** is replaced with the overmolded insert **400**. In another example, the subsonic cartridge casing **600** can be formed from two pieces, an upper and lower component similar to that described above. However, the components have a constant second diameter **630** between the two. The lower component can be formed either with the insert or without and the back end **614** is polymer.

FIGS. 19A and 19B illustrate a further example of a subsonic cartridge **700**. In this example, a full metal sleeve **702** extends a significant length of the cartridge **700**. The sleeve **702** can have an insert section **704** similar to the insert **400**, and the sleeve **702** can act as an integral extension of the insert **400**. The insert section **704** can have a self-reinforced area **714** which can include an extraction groove **705** and a rim **706**. The groove **705** and rim **706** are dimensioned to the specific size as dictated by the caliber of the ammunition. The insert section can also have a primer pocket **716** and flash hole **718**.

Forward of the insert section **704** is sleeve section **708**. The sleeve section **708** can extend the length of the cartridge **700** and, in one example, form a neck **710** of the cartridge with a mouth **712** wherein the bullet **50** is fitted into the mouth **712**. The mouth **712** can have a mouth diameter **720** sized to receive the bullet **50** and the remaining portion of the sleeve section **708** can have a sleeve diameter **722** approximately equal to the mouth diameter **720**. The sleeve section **708** can act as a propellant chamber **724**, and the sleeve diameter **722** can be such as to limit the amount of propellant so the bullet **50** can travel at subsonic speeds.

In an example, the sleeve **702** is straight walled and the sleeve diameter **722** approximates a bullet diameter **51**. To allow the cartridge **700** to fit in a standard chamber for the

particular caliber, the outside of the sleeve **702** is molded with a polymer sheath **800**. The polymer sheath **800** can be molded to the true dimensions of the cartridge for the particular caliber, including a shoulder **802** and outside wall **804**. Multiple ridges **726** can be formed in the sleeve section **708** to allow the polymer from the polymer sheath **800**, during molding, to form bands (not illustrated and as above). The combination of the ridges **726** and bands aid in resisting separation between the sleeve **700** and the polymer sheath **800**. The resistance can be most important during the extraction of the cartridge from the firearm by an extractor (not illustrated).

The ridges **726** can also include one or more keys **728**. The keys **728** are flat surfaces on the ridges **726**. The keys **728** prevent the sleeve **702** and the polymer sheath **800** from rotating in relation to one another, i.e. the sleeve **702** twisting around in the polymer sheath **800**. Instead of, or in conjunction with, the ridges **726**, the sleeve **702** can have knurling or texturing **730** to prevent the rotational rotation.

In other examples, the sleeve section **702** does not extend the length of the cartridge **700**. The sleeve section **702** can stop at or before the molded shoulder **802**. In this example, the polymer sheath **800** can form a polymer neck **806** and polymer mouth **808** to receive the bullet **50**. See FIG. 19C.

The sleeve **702** can be metal and formed by turning down bar stock to the specific dimensions or can be cold formed. Further, it can be a different metal than the insert section **704**. The goal is to create a lightweight cartridge using the strength of the metal sleeve and the low weight, high strength properties of polymers. Using more polymer than metal assists in the weight to strength ratio. The polymer sheath **800** can be made of the same polymers discussed above or other polymers of lower strength, owing to the metallic support of the sleeve **702**. The metals can be any known metals that can provide light weight strength under exploding propellant conditions. This includes brass, aluminum, steel or other alloys. Further, ceramics or other materials may also be used.

In one example, the sleeve **702** can be a brass cartridge from a different caliber (typically smaller) that receives a polymer sheath to fit in a larger caliber chamber. The brass cartridge can also be cut or stretched to accommodate the larger caliber bullet and the particular length required of the cartridge. Note that in a further example, the sleeve **702** can have sloped shoulders and the shoulders can remain exposed or sheathed in polymer. In other examples, the insert section **704** and the sleeve section **708** are not integral. They can be separated and molded as one piece, as in FIG. 20A. Alternately, the examples above can have a lower component **900** of polymer **902** and the insert section **704** polymer welded to an upper component **904** of polymer and sleeve section **708**. The upper and lower components **900**, **904** can have a mating overlap/underskirt/taper section **906**, as described above. Either component **900**, **904** can have an overlap or underskirt portion and the opposite component **900**, **904** can have the mating taper portion. See FIG. 20B. The lower and upper components **900**, **904** can be similar to the lower and upper components described above in assembly and size. FIG. 20C illustrates the sleeve **702** without the insert section **704**, only the sleeve section **708**. In this example, the polymer sheath **800** forms a back end **814**, similar to the polymer back end **614** described above.

Additional examples of reduced capacity cartridge cases are illustrated in FIGS. 21 and 22. FIG. 21 illustrates a lower narrowed cartridge **1000**. The lower narrowed cartridge **1000** includes an upper component **1200** of the lower narrowed cartridge, a lower component **1300** of the lower

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narrowed cartridge and an insert **1400** for the lower narrowed cartridge. The upper, lower, and insert **1200**, **1300**, **1400** are generally formed as above, except as described further below. The upper component **1200** has a mouth **1208** in which a bullet **1050** is inserted. The mouth **1208** is an opening in the neck **1206** of the upper component **1200** and can also contain a lip **1214**. The lip **1214** can engage a cannellure **1055** in the bullet **1050**.

Further, at least one the lip **1214** and the cannellure **1055** can be replaced with an adhesive (not illustrated). The adhesive can seal the bullet **1050** in the neck **1206** and provide a waterproofing feature, to prevent moisture from entering between the bullet **1050** and the neck **1206**. The adhesive also provides for a control for the amount of force required to project the bullet **1050** out of the cartridge **1000**. Controlling this exit force, in certain examples, can be important, since the bullet for sub-sonic ammunition is already "under powered" in relation to a standard round.

The bullet **1050** is a standard weight bullet for its particular caliber. The "standard weight" or common weight for a projectile varies slightly. Some examples of standard weights can include at .223 (5.56) caliber weights between 52 and 90 grains; at .308 and .300 Winchester Magnum calibers weights between 125 and 250 grains; and for .338 Lapua® Magnum caliber weights between 215 and 300 grains. This can also include standards weights for .50 caliber between 606 and 822 grains. The bullet **1050** can be less than 125% of maximum standard weight for a particular caliber. Further, the bullet can be less than 120%, 115%, 110% and 105% of the caliber's maximum standard weight.

The upper component **1200** can also include a shoulder **1204**. The shoulder **1204** slopes outward from the neck **1206** and then straightens out to form the upper component outer wall **1217**. The upper component **2100** can join the lower component **1300** as described above, and the lower component **1300** also can have a lower component outer wall **1317**. The upper and lower component outer walls **1217**, **1317** can form the outer shape of the cartridge and are shaped as such to fit a standard chamber for the particular caliber.

Both the upper and lower components **1200**, **1300** can have inner walls **1219**, **1319**, respectively. The inner walls **1219**, **1319** can form the propellant chamber **1340**, which contains the powder or other propellant to discharge the bullet **1050** from the weapon (not illustrated). The inner walls **1219**, **1319**, in this example, can be angled to form a constant slope toward the insert **1400**. This narrows, or tapers, the propellant chamber **1340** so the diameter D1 in the upper component **1200** is greater than the diameter D2 closer to the insert **1400**. It can be further said that, in an example, a diameter D1 approximate the shoulder **1204** can be greater than the diameter D2 (in the lower component **1300**) approximate a flash hole **1418** of the insert **1400**. In another example, diameter D2 can equal a diameter D3 of the flash hole **1418**.

FIG. 22 illustrates another example of a narrowed propellant chamber **1340**. In this example, the propellant chamber **1340** narrows toward the upper component **1200**. Thus, a diameter D4 of the upper component **1200** is less than a diameter D5 of the lower component **1300**. Additionally, the diameter of the lower component D5 can be greater than the diameter D3 of the flash hole **1418**. In one example, the diameter D4 of the upper component **1200** is greater than or equal to a diameter D6 of a back of the bullet **1050**.

In the above examples, the cartridge **1000** is described in a three-piece design (upper **1200**, lower **1300**, and insert **1400**). Note that the cartridge **1000** can be fabricated in one-piece, all of polymer as described above, or two pieces,

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a polymer section and the over-molded insert **1400**. Additionally, the flash hole **1418** can also be sloped to match the slope of the inner walls **1217**, **1317**. Further, while the above examples are described with a constant slope from the upper component **1200** to the lower component **1300**, other examples can have differing slopes between the two components **1200**, **1300** such that one slope is steeper than the other slope. Further, FIGS. 21 and 22 illustrate cartridges wherein the upper component **1200** is smaller than the lower component **1300**. The relative sizes of the two components **1200**, **1300**, can be alternated or they can be equated.

Further, the slope of the upper component inner wall **1219** can differ from the upper component outer wall **1217**. The same can be true for the lower component inner wall **1319** differing in slope from the lower component outer wall **1317**.

The polymer construction of the cartridge case also provides a feature of reduced friction between the cartridge and chamber of the firearm. Reduced friction leads to reduced wear on the chamber, further extending its service life.

Subsonic ammunition can be manufactured using the above illustrated examples. Subsonic ammunition is designed to keep the bullet from breaking the speed of sound (approximately 340 m/s at sea level or less than 1,100 fps). Breaking the speed of sound results in the loud "crack" of a sonic boom, thus subsonic ammunition is much quieter than is standard counterpart. Typical subsonic ammunition uses less powder, to produce less energy, in the same cartridge case as standard ammunition. The remaining space is packed with wadding/filler to keep the powder near the flash hole so it can be ignited by the primer. As noted above, increasing the wall thickness eliminates the need for wadding. In one example, while a brass cartridge wall can be 0.0389" thick, the polymer wall and sleeve can have a total thickness of 0.0879" for the identical caliber.

The reduced capacity allows for a more efficient ignition of the powder and a higher load density with less powder. Low load density (roughly below 30-40%) is one of the main contributors to the Secondary Explosive Effect (SEE). SEE can destroy the strongest rifle action and it can happen on the first shot or the tenth. SEE is the result of slow or incomplete ignition of small amounts of smokeless powder. The powder smolders and releases explosive gases which, when finally ignited, detonate in a high order explosion. The better sealing effect is also important here because standard brass does not seal the chamber well at the lower pressures created during subsonic shooting.

While the foregoing has described what are considered to be the best mode and/or other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that the teachings may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all applications, modifications and variations that fall within the true scope of the present teachings.

What is claimed is:

1. A high strength polymer-based blank cartridge comprising:

- a first end;
- a shoulder disposed approximate the first end;
- a second end opposite the first end;
- an insert engaged to the second end;
- a metal sleeve integral with the insert, extending continuously from the insert to the shoulder;

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- a polymer sheath covering at least the metal sleeve, the polymer sheath molded directly over an outer surface of the metal sleeve; and
- a propellant chamber, formed between the shoulder and the insert by at least the metal sleeve, and having an unobstructed and a reduced volume; 5
- the reduced volume is reduced in comparison to a propellant chamber of a standard cartridge.
2. A high strength polymer-based blank cartridge of claim 1, further comprising: 10
- a mouth, formed in the metal sleeve, disposed on the first end, opposite the insert.
3. A high strength polymer-based blank cartridge of claim 2, wherein the polymer sheath extends between the insert and the mouth. 15
4. A high strength polymer-based blank cartridge of claim 2, wherein the polymer sheath extends between the insert and the shoulder.
5. A high strength polymer-based blank cartridge of claim 1, wherein the metal sleeve is straight-walled. 20
6. A high strength polymer-based blank cartridge of claim 1, wherein the metal sleeve is integral with the insert.
7. A high strength polymer-based blank cartridge of claim 1, wherein the metal sleeve comprises knurling.

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